Inequality Aversion and the Measurement of Real Income

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Abstract

Under non-homothetic preferences, the measurement of real income is not ethically innocuous. Money-metric utility and related welfare indices require a choice of reference prices, and different admissible choices can change conclusions about growth comparisons, inflation inequality, and aggregate welfare. This paper studies how individual welfare should be cardinalized, and how social welfare should be evaluated, when that normalization choice itself affects the degree of inequality aversion embodied in the social ranking. The paper proposes an axiomatic approach in which inequality aversion is made primitive through a transfer principle based on tolerated leakage in equalizing income transfers. The tolerated leakage is summarized by a concave function ϕ, yielding the notion of ϕ-prioritarianism. Together with standard axioms, it characterizes a canonical notion of real income and a canonical social ranking. Under additional conditions, the latter can be written as a prioritarian sum of money-metric utilities with an endogenously selected reference price system. The paper then derives explicit formulas for standard non-homothetic demand systems, including Gorman polar form, PIGL, and non-homothetic CES, and shows how the resulting welfare measure can be expressed as a correction to Divisia-style growth indices. Finally, the analysis is extended to heterogeneous preferences, clarifying what can and cannot be recovered once interpersonal comparability must also be addressed.

Keywords: Real income, Inequality aversion, Money-metric utility, Non-homotheticity, Prioritarianism, Interpersonal comparisons.

JEL codes: D60, D63, C43, D31, E31.

1 Introduction

Under homothetic and identical preferences, the choice of reference prices in money-metric utility is largely innocuous. Under identical but non-homothetic preferences, it is not. The same economy can exhibit different measured real-income growth and different measured inflation inequality, depending only on the price vector used to cardinalize utility. This paper thus asks how real income should be measured, and more generally how social welfare should be evaluated, when that normalization choice is itself ethically consequential.

A growing empirical literature shows that income effects matter both for aggregate real-income growth measurement (Comin, Lashkari, and Mestieri 2021; Baqaee and A. Burstein 2023) and for the distributional incidence of inflation (Jaravel 2021; Jaravel and Lashkari 2024). The canonical welfare-theoretic tools for addressing those questions are money-metric utility and exact cost-of-living indices (Konüs 1939; McKenzie 1957; Samuelson and Swamy 1974). Empirical work accordingly tends to treat money-metric utility as the natural benchmark for real income. The difficulty is that the reference prices entering the money-metric are rarely selected by the theory itself. In practice they are often fixed at an initial date, a final date, or some intermediate observed price vector, with little normative reason to prefer one admissible reference system to another.

That arbitrariness matters for at least three reasons. First, moving reference prices can generate dynamically inconsistent welfare comparisons: one may record strictly positive measured growth period after period and nevertheless return to the initial situation (Fleurbaey and Blanchet 2013). Second, some reference-price choices yield welfare representations that are not concave, opening the door to anti-egalitarian implications (Blackorby and Donaldson 1988). Third, and most importantly for applied work, different plausible reference prices can reverse substantive conclusions - as the motivating section 2 illustrates. The same data may suggest that standard growth statistics understate welfare growth under one reference system and overstate it under another; likewise, the same price changes may appear to hurt poorer households more under one normalization and richer households more under another. What, then, is the welfare conclusion?

One ethical issue raised by reference-price choice in money-metric utility is its effect on interpersonal comparisons under heterogeneous preferences: which individual counts as worst off may depend on the reference prices used. This paper is concerned with a distinct issue. Even under identical preferences, reference-price choice still matters ethically because, when combined with a concave social aggregator, it shapes the degree of inequality aversion encoded in the social ranking. In such a setting, money-metric utility is not primarily an instrument for interpersonal comparability, but a device for welfare normalization. The relevant question is then: abstracting from interpersonal comparability, how should welfare cardinalization and inequality aversion be consistently articulated? In the case of infinite inequality aversion, the answer is known: one is driven to a Rawlsian criterion. For finite inequality aversion, however, the issue remains largely unresolved. The literature has shown that minimal fairness requirements of the Pigou-Dalton type impose the use of concave welfare representations in social evaluation (Gorman 1959; Fleurbaey and Tadenuma 2014; Piacquadio 2017). But it does not say how a strictly positive finite degree of inequality aversion should constrain the choice of welfare cardinalization itself.

The paper's proposal is to make inequality aversion primitive in the axioms rather than implicit in the normalization. A useful way to see the approach is as an extension of the familiar idea of a least concave representation of preferences (Debreu 1976). Under the standard Pigou-Dalton principle, a minimally inequality-averse society would select, among the social rankings that respect Pareto and endorse every non-leaky equalizing transfer, one that is as weakly inequality-averse as possible. In the identical-preference case, this - along with usual basic axioms such as separability - leads to the use of least concave utility representations: among all concave welfare cardinalizations compatible with the transfer principle, one selects the one that imposes the smallest amount of curvature beyond what fairness requires. The present paper generalizes that logic to arbitrary finite degrees of inequality aversion. Instead of requiring only that every non-leaky progressive transfer in resources improve social welfare, the axioms fix how much leakage society is willing to tolerate in order to achieve more equality. That tolerated leakage is summarized by a concave function ϕ. When both individuals face the same prices, a transfer from a richer individual to a poorer one is required to be welfare-improving whenever the poorer individual's gain in income and the richer individual's loss exactly offset in ϕ-space. The curvature of ϕ therefore directly encodes the strength of prioritarian concern: a more concave ϕ allows society to endorse more leaky equalizing transfers.

This leads to the notion of ϕ-prioritarianism. A social ranking is ϕ-prioritarian if it endorses all transfers prescribed by the above ϕ-transfer criterion and, among all rankings that do so, is the least inequality-averse one - where a ranking is less inequality-averse than another if every equalizing income transfer that the former endorses is also endorsed by the latter. In that sense, ϕ-prioritarianism extends the least-concavity idea: when ϕ is linear, one recovers the benchmark in which only non-leaky transfers must be accepted, and the relevant welfare cardinalization is the least concave one; when ϕ is strictly concave, one instead looks for the least inequality-averse welfare cardinalization compatible with a richer set of admissible leaky transfers. The ethical content of inequality aversion is thus fixed directly at the axiomatic level, rather than being left to emerge indirectly from an arbitrary normalization of utility.

In the identical-preference case, ϕ-prioritarianism, together with the standard axioms of continuity, separability and Pareto, pins down a unique social ranking whenever the relevant regularity conditions hold. The characterization is constructive. For a given degree of prioritarian concern, the axioms identify a canonical transformation of utility, hence a canonical notion of real income. Under additional assumptions on the type of non-homotheticity of the underlying preferences, this real-income representation can in turn be expressed in money-metric form. In that case, there exists a reference price system $\tilde{p}$ such that canonical real income is represented by the money-metric utility evaluated at $\tilde{p}$. The characterized social ranking can then be written as a prioritarian sum of money-metric utilities defined with an endogenous reference price.

The framework is also constructive in a more applied sense. It yields closed-form results for familiar non-homothetic demand systems, including Gorman polar forms, PIGL specifications, and non-homothetic CES. In each case, the paper shows how the ethically selected real-income representation can be written as a correction to a Divisia-style growth measure. This matters for current empirical debates. This paper provides a normative foundation for the benchmark against which Divisia measures should be assessed.

Finally, the paper returns to the issue that was set aside at the outset: heterogeneous preferences. The last part of the analysis shows that the same logic can be extended beyond the common-preference benchmark, once one is explicit about the interpersonal criterion used to compare individuals with different preferences. The paper studies both a unique-profile framework and a multi-profile framework. In each case, ϕ-prioritarianism can be combined with suitable interpersonal fairness axioms to characterize social rankings whenever the admissible preference domain allows it. ϕ-prioritarianism does not by itself solve the interpersonal comparability problem: it tells us how to cardinalize and aggregate well-being once an interpersonal measure has been chosen, but it does not select that measure.

The contribution is therefore both normative and applied. Normatively, the paper contributes to the welfare-economic literature on money-metric and opportunity-equivalent representations of well-being (Samuelson and Swamy 1974; Pazner 1979; Thomson 1994; Fleurbaey 2009; Fleurbaey and Maniquet 2011; Fleurbaey and Tadenuma 2014; Piacquadio 2017; Bosmans, Decancq, and Ooghe 2018; Schlee and Khan 2023). Whereas this literature is mainly concerned with interpersonal comparability under heterogeneous preferences, the present paper isolates an indeterminacy that remains even under identical preferences: the interaction between welfare cardinalization and finite inequality aversion. The paper also contributes to the literature on the concavification of utility representations. In the benchmark case of minimal inequality aversion, the characterization yields a new formulation of the least-concave representation of preferences, closely related to the classical construction of Kannai (1977). The present approach extends that logic to any finite degree of prioritarian concern. From an applied perspective, the paper connects to the literature on growth and inflation mismeasurement under income effects (Crawford and Oldfield 2002; Jaravel 2021; Jaravel and Lashkari 2024; Baqaee, A. T. Burstein, and Koike-Mori 2024; Chen, Levell, and O'Connell 2024), and taste shocks (Fisher and Shell 1972; Baqaee and A. Burstein 2023) by providing a normatively grounded benchmark against which the biases of conventional statistical measures can be evaluated. It also relates to the literature on optimal taxation and redistribution in the presence of income effects (Ferriere, Grubener, and Sachs 2024), by proposing a social objective from which optimal policy prescriptions can be derived.

The rest of the paper is organized as follows. Section 2 presents three motivating examples showing how reference-price choice can reverse conclusions about inflation inequality, long-run growth, and aggregate welfare. Section 3 introduces the axiomatic framework and the characterization results in the identical-preference case. Section 4 derives transparent formulas under standard demand systems and presents a modest empirical illustration for non-homothetic CES. Section 5 extends the analysis to heterogeneous preferences. Section 6 concludes.

2 Motivating examples

The purpose of this section is to isolate the problem identified in the introduction, namely that different plausible choices of reference prices can reverse substantive conclusions in applied welfare analyses. We first introduce the common objects used in such analyses.

Real income and inflation in empirical welfare analyses. Consider a household whose utility evolves from u_0 to u_t as prices change from p_0 to p_t and nominal income from I_0 to I_t.

Let $e(p, u)$ denote the expenditure function associated with the household's preferences. Real income growth and inflation inequality measures are usually based on money-metric utility functions, defined at date t for any reference price $\tilde{p}$ as

$$M_{\tilde{p}}(u_t) := e(\tilde{p}, u_t).$$

Real income growth measured in this reference system is therefore

$$\frac{M_{\tilde{p}}(u_t)}{M_{\tilde{p}}(u_0)} = \frac{e(\tilde{p}, u_t)}{e(\tilde{p}, u_0)}$$

and inflation $\Pi_{\tilde{p}}$ is given by

$$\Pi_{\tilde{p}}(p_0, p_t, I_0, I_t) := \frac{I_t/I_0}{e(\tilde{p}, u_t)/e(\tilde{p}, u_0)}$$

so that it acts as a deflator to nominal income growth, since one then has

$$\frac{M_{\tilde{p}}(u_t)}{M_{\tilde{p}}(u_0)} = \frac{I_t}{I_0}/\Pi_{\tilde{p}}.$$

In practice, the reference price selection usually boils down to choosing between two alternatives: initial prices p_0 and final prices p_t. Inflation $\Pi_{\tilde{p}}$ then simplifies to

$$\Pi_{p_0} = \frac{e(p_t, u_t)}{e(p_0, u_t)}, \qquad \Pi_{p_t} = \frac{e(p_t, u_0)}{e(p_0, u_0)}. \tag{1}$$

where the right-hand-sides of the equalities correspond to cost-of-living indices (Konüs 1939). For these specific reference price choices, inflation can therefore be defined without explicit reference to money-metric utility.

In three deliberately stylized environments, even these two natural candidates - initial and final prices - can point in opposite directions. In the first example, they reverse the ranking of inflation across poor and rich households. In the second, Divisia growth lies between two money-metric paths, so the sign of the alleged bias depends entirely on the normalization. In the third, the same policy is socially improving under one money-metric system and socially worsening under another.

2.1 Inflation Inequality and Reference Prices

The fact that income effects give rise to heterogeneous inflation rates among different income groups is well established. For that reason, analyses of inflation inequality typically estimate separate money-metric growth rates for different income groups. However, once expenditure patterns depend on income, the claim that one group experienced more inflation than another is not invariant to the choice of price system used to evaluate welfare changes. We illustrate this point with a simple simulation based on the non-homothetic CES specification of Comin, Lashkari, and Mestieri (2021), which generates Engel curves consistent with structural transformation and in which non-homotheticity persists even at high income levels.

Setup. Households share identical preferences represented by the nh-CES expenditure function

$$e(p,u) = \left(\sum_{i=1}^{n} \Omega_i \left(u^{\epsilon_i} p_i\right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}},$$

where u is utility, p is the price vector, σ is the elasticity-of-substitution parameter, Ω is a vector of share parameters and ϵ_i governs the elasticity of demand for good i. Following Comin, Lashkari, and Mestieri (2021), we work with three sectors (agriculture, manufacturing, and services).

Consider two households, labeled poor and rich, facing the same sequence of sectoral prices and the same proportional growth in nominal income, but starting from different income levels. We simulate a scenario where the relative price of manufacturing rises over time, while the other two sectoral prices remain stable. Figure 1 shows the evolution over time of the 'true' price index $\pi_{\tilde{p}}$ for both households, for two different reference prices - initial and final prices.

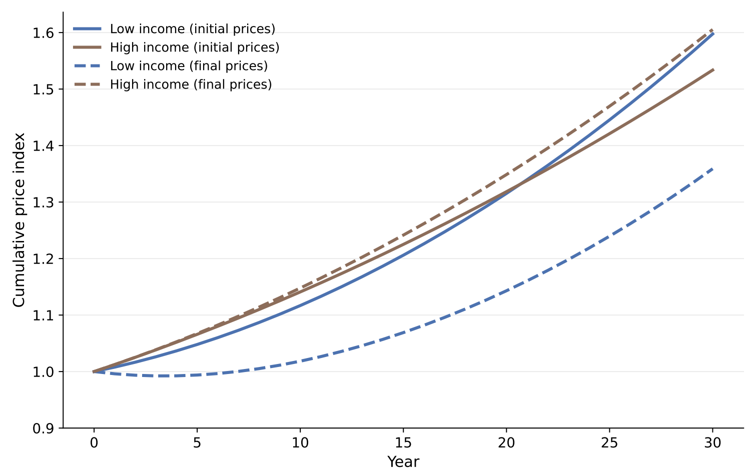

Figure 1: Cost of living indices for poor and rich households under initial and final price reference systems

The blue solid line ends up above the brown solid line: with initial prices as reference, total measured inflation favors the rich. The opposite is true with final prices.

Parameters: calibrated as in Comin, Lashkari, and Mestieri (2021). $\sigma = 0.26$, $(\varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.2, 1, 1.65)$.
Prices: (p_a, p_m, p_s) go from $(1,1,1)$ to $(1,3,1)$

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With initial prices as reference, total inflation over the whole period is such that there is inflation inequality favoring the rich, while the opposite is true for final prices as reference. In other words, the direction of inflation inequality reverses when one uses final prices instead of initial prices as reference.

Mechanism. Denote by $u_{P,0}$ and $u_{P,t}$ the initial and final utility levels of the poor household, and $u_{R,0}$ and $u_{R,t}$ the utility levels of the rich household. Let $\pi_{\tilde{p}}^{P}$ be the measured inflation for the poor household (between 0 and $t = 30$), and $\pi_{\tilde{p}}^{R}$ be inflation for the rich household.

By definition, the direction of inflation inequality depends on the sign of

$$\pi_{\tilde{p}}^{P} - \pi_{\tilde{p}}^{R}.$$

Let $\Gamma : u \mapsto \frac{e(p_t,u)}{e(p_0,u)}$ denote the cost-of-living index as a function of utility, for fixed initial and final prices. Considering the equivalence result shown by equation 1, inflation inequality reversal (for reference prices equal to initial prices vs. final prices) thus happens whenever

$$\Gamma(u_{R,T}) - \Gamma(u_{P,T}) \quad \text{and} \quad \Gamma(u_{R,0}) - \Gamma(u_{P,0})$$

have opposite signs. If $u \mapsto \Gamma(u)$ is a monotone function of u, this reversal of signs cannot happen as long as the rich household stays richer than the poor household throughout the period. But, denoting $w(p,u)$ the vector of Hicksian expenditure shares,

$$\Gamma(u) = \exp \int_0^t \sum_i w_i(p(r),u)\frac{\dot{p}_i}{p_i}(r)dr$$

which, in our example where only the price of manufacturing changes, simplifies to

$$\Gamma(u) = \exp \int_0^t w_m(p(r),u)\frac{\dot{p}_m}{p_m}(r)dr$$

$\Gamma(u)$ increases (decreases) with utility if individuals consume relatively more (less) of goods whose prices have increased over time. But if budget shares exhibit non monotone patterns as a function of utility, $\Gamma(u)$ need not be monotone. This non-monotonicity of Engel curves is an empirically relevant feature and in fact a strength of the nh-CES specification which allows for such a property. The hump-shape of the manufacturing Hicksian share as a function of utility - illustrated in figure 2 - is the mechanism behind the reversal of the direction of measured inflation inequality. In the initial period, the rich household devotes relatively more income to the manufacturing sector, but the opposite is true at the final period.

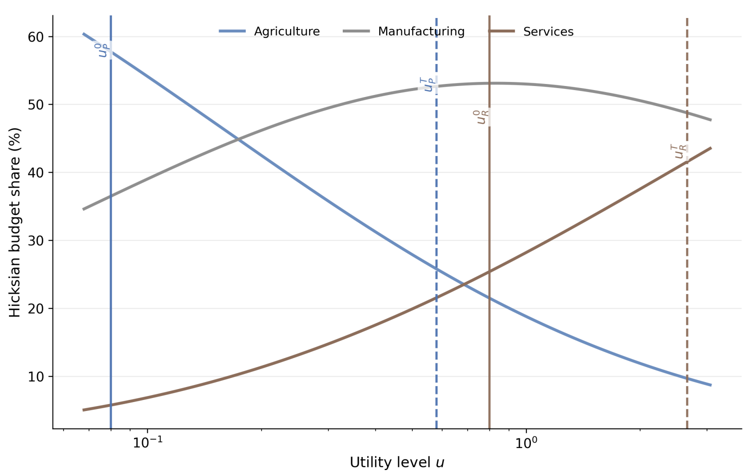

Figure 2: Hicksian budget shares (at final prices in this graph). Their variation with utility governs the slope of the cost-of-living profile $\Pi(u)$.

The key take-away of this first example is that, under non-homothetic preferences, inflation inequality is an object that depends on the reference prices used to convert price and quantity changes into real-income measures.

2.2 Long-run growth

A closely related issue arises in long-run growth measurement via Divisia indices, the continuous-time counterpart of the chained indices used by national statistics agencies.

Definition 2.1 (Divisia indices)**.** Let $x_t = (x_{1t}, \ldots, x_{nt})$ denote the consumption vector at date t, and let $p_t = (p_{1t}, \ldots, p_{nt})$ denote the corresponding price vector. Define expenditure shares

$$s_{it} = \frac{p_{it}x_{it}}{\sum_{j=1}^{n} p_{jt}x_{jt}}.$$

The Divisia quantity index between dates 0 and T is

$$\log \frac{Q_T}{Q_0} = \int_0^T \sum_{i=1}^{n} s_{it}\, d\log x_{it}.$$

The Divisia price index between dates 0 and T is

$$\log \frac{P_T}{P_0} = \log \frac{I_T}{I_0} - \int_0^T \sum_{i=1}^{n} s_{it}\, d\log x_{it} = \int_0^T \sum_{i=1}^{n} s_{it}\, d\log p_{it}$$

Applied work often compares Divisia quantity growth with money-metric growth and interprets the difference as a bias. Yet the sign of this bias depends on the reference price. Using the same simulation as above, Figure 3 plots cumulative growth for the low-income household. In this example, the Divisia path lies between the two money-metric growth paths generated by fixing reference prices at the initial or final date.

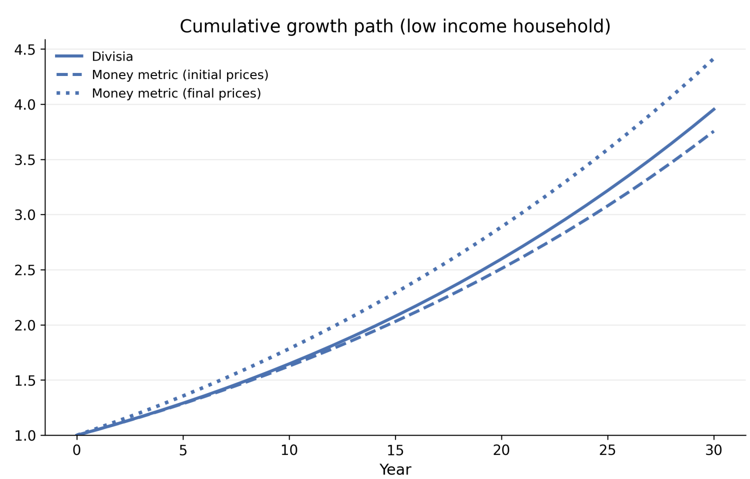

Figure 3: Cumulative growth of the low-income household: Divisia versus money-metric indices under initial and final reference prices.

Conclusions on potential growth mismeasurement by Divisia indices may become vacuous if there is no reason to favor one reference price over another. In fact, Divisia growth is itself a welfare representation whenever the observed consumption path $t \mapsto x_t$ is such that utility is strictly monotone over time. Let $u_t = U(x_t)$, and let $t(u)$ denote the inverse along the path (time as a function of utility, which is well defined because of the monotonicity assumption on utility over time). Define the utility-indexed price schedule

$$\tilde{p}(u) = p\big(t(u)\big).$$

Then Divisia quantity growth between dates 0 and T can be written as

$$\exp \int_{u_0}^{u_t} \frac{\partial \log e(\tilde{p}, u)}{\partial \log u}\bigg|_{\tilde{p}=\tilde{p}(u)} d\log u.$$

This expression is a strictly increasing transform of utility, hence a valid welfare representation. Moreover, it is naturally interpreted as a money-metric utility index with moving reference prices. For a fixed reference price $\bar{p}$, money-metric utility is $M_{\bar{p}}(u) = e(\bar{p}, u)$, and its growth is given by

$$\frac{M_{\bar{p}}(u_t)}{M_{\bar{p}}(u_0)} = \exp \int_{u_0}^{u_t} \frac{\partial \log e(\bar{p}, u)}{\partial \log u} d\log u.$$

Divisia growth differs only in replacing the fixed reference price $\bar{p}$ by the utility-indexed price schedule $u \mapsto \tilde{p}(u)$. It is therefore the money-metric differential evaluated at a reference price system that moves along the realized path.

2.3 Social welfare evaluation

Finally, the reference-price issue becomes normatively acute once individual welfare changes are aggregated. Consider a social evaluator who must decide whether to endorse a policy that raises the price of one good and lowers the price of another. Even in a very simple environment, the social ranking can depend on the chosen money-metric normalization.

Consider two consumers, indexed by $i \in \{P, R\}$ (poor and rich), two goods, and two periods. Good 1 is an essential good and good 2 is a non-essential good. Preferences are Stone-Geary:

$$U(x_1, x_2) = (x_1 - \gamma_1)^{\alpha_1} x_2^{\alpha_2},$$

with $\gamma_1 > 0$ denoting the subsistence requirement for the essential good. The prices of the two goods are equal in the first period, but the essential good becomes more expensive in the second period, while the non-essential good becomes cheaper. Let $M_{i,\tilde{p}}$ denote money-metric utility of consumer i, and assume aggregate welfare is evaluated with a log-utilitarian criterion,

$$W_t(\tilde{p}) = \exp\left(\frac{1}{2} \sum_{i \in \{P,R\}} \log M_{i,\tilde{p}}(u_{it})\right).$$

Figure 4 shows that changing the reference price vector reverses the sign of the aggregate welfare assessment. The same underlying allocation path is judged welfare-improving under one money-metric normalization and welfare-reducing under another.

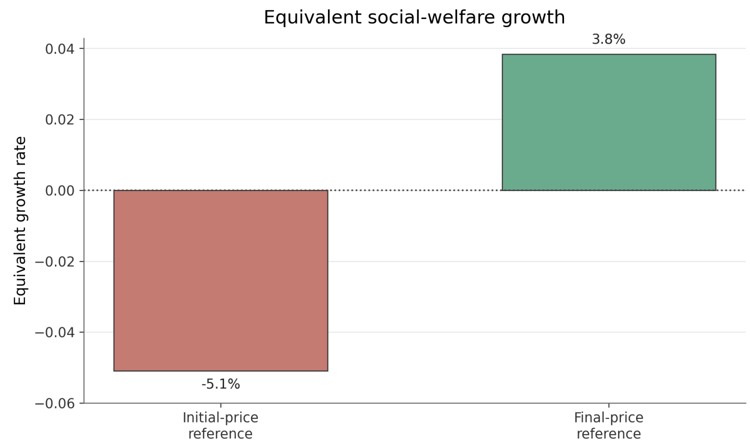

Figure 4: Equivalent aggregate welfare growth under two reference-price systems. A change in the money-metric reference price is enough to reverse the welfare verdict.
Parameters: $p_0 = (1, 1)$, $p_1 = (1.8, 0.25)$, $(I_P, I_R) = (10, 50)$, $\gamma_1 = 5$, $\alpha_1 = \alpha_2 = 0.5$.

Mechanism. If preferences were homothetic ($\gamma_1 = 0$), both consumers would benefit from the price changes since both goods enter symmetrically in the utility function ($\alpha_1 = \alpha_2$) and one good's price is multiplied by less than 2 while the other's price is divided by almost 4. But because of the positive subsistence level γ_1, the poor household still prefers the initial prices because their utility strongly relies on the necessity good x_1. Thus total welfare change is the sum of a negative term (the welfare change of the poor household) and a positive term (the welfare change of the rich household), and the sign of total welfare change depends on the magnitude of these terms. For our Stone-Geary preferences, money-metric utility is given by

$$M_{i,\tilde{p}}(u_{it}) = \tilde{p}_1\gamma_1 + u_{it}\left(\frac{\tilde{p}_1}{\alpha_1}\right)^{\alpha_1}\left(\frac{\tilde{p}_2}{\alpha_2}\right)^{\alpha_2}$$

For high utility levels, the constant term $\tilde{p}_1\gamma_1$ is small in comparison with the linear term in u, so the magnitude of the welfare change of the rich household - $\Delta \log M_{i,\tilde{p}}(u_R)$ - is almost insensitive to $\tilde{p}$. For the poor household, however, $\Delta \log M_{i,\tilde{p}}(u_P)$ will be substantially larger (i.e. very negative) whenever $\tilde{p}_1\gamma_1$ is low, because relative growth becomes very large for very small denominators. That is why total welfare change is driven by the welfare change of the poor household when the reference price is such that the price of the essential good is low, which corresponds to initial prices in our simulation.

Taken together, the three examples sharpen the paper's central motivation. Under non-homothetic preferences, reference prices do not merely index purchasing power. They also select a cardinal welfare representation and, once combined with a concave social aggregator, shape the effective degree of inequality aversion. Section 3 addresses that problem directly. Rather than treating the welfare normalization as an arbitrary preliminary step, it makes the relevant distributive judgment explicit through an axiom on admissible transfers. The goal is to recover a social ordering in which the degree of inequality aversion is fixed first, and the corresponding notion of real income is then derived from it.

3 Real income under ϕ-prioritarianism

3.1 Framework

We consider a population of N consumers with preferences over bundles $x \in X = \mathbb{R}^L_+$.[1] Throughout this section and the next section, preferences are common across individuals. We assume that preferences can be represented by a continuous, strictly increasing, and concave utility function. Let $U : X \to \mathbb{R}_{++}$ be such a representation. Each consumer i is endowed with a budget set $B_i = (p_i, I_i)$, with $p_i \in \mathbb{R}^L_{++}$ and $I_i > 0$. Let $\Delta = \{p \in \mathbb{R}^L_{++}, \sum_l p_l = 1\}$ be a set of normalized prices. Denote by

$$V(p, I) = \max\{U(x) : px \leq I\}$$

the indirect utility function and by

$$e(p, u) = \min\{px : U(x) \geq u\}$$

the expenditure function. For each budget set $B_i = (p_i, I_i)$, let $x^*(B_i) = x^*(p_i, I_i)$ denote the associated Marshallian demand.

A social ordering $\succsim$ is a weak order over allocations of bundles $\mathbf{x} = (x_1, ..., x_N)$. Alternatively, one may think of $\succsim$ as an order over allocations of budget sets $\mathbf{B} = ((p_i, I_i))_{i=1}^N$ through the induced bundle profile $x^*(B) = (x^*(B_1), \ldots, x^*(B_N))$. Whenever the social ranking over allocations is represented by a social welfare function $W(x)$, we will abuse notation by writing $W(B)$ for the indirect social welfare of an allocation of budget sets $B = ((p_i, I_i))_i$, defined by

$$W(B) = W\big((x^*(B_i))_i\big).$$

For any pair of allocations x, x', $x \succsim x'$ means that society considers x to be at least as desirable as x', and $x \simeq x'$ means that society is indifferent between the two allocations. The goal of this section is to axiomatically pin down a social ranking from four basic axioms and a novel axiom called ϕ-prioritarianism. All proofs are in Appendix B.

3.2 Basic axioms

The first three axioms are standard in the literature and induce an additive representation of social welfare. The first one - continuity - ensures that small perturbations of individual bundles do not lead to discontinuous changes in the social ranking.

Axiom 3.1 (Continuity). *For each $x \in X^N$, the sets $\{x', x \succsim x'\}$ and $\{x', x' \succsim x\}$ are closed.*

Next, separability says that the social comparison of two allocations should not depend on the bundles given to unconcerned individuals (those that receive the same bundle in both allocations). In the present context it is the key axiom behind additivity.

[1] Nothing essential hinges on taking the whole commodity space as the consumption set: X could instead be any suitable subset of $\mathbb{R}^L_+$, provided it satisfies the topological and convexity conditions needed for the axiomatic results below.

Axiom 3.2 (Separability). *For every subset of individuals $S \subseteq \{1, \ldots, N\}$, for any bundle profiles x_S, y_S, and for any two complements z_{-S}, z'_{-S},*

$$(x_S, z_{-S}) \succsim (y_S, z_{-S}) \iff (x_S, z'_{-S}) \succsim (y_S, z'_{-S}).$$

Pareto guarantees that the social criterion respects unanimous improvements according to the common preference ordering. It is the minimal requirement ensuring that the social ordering is increasing in each individual's well-being.

Axiom 3.3 (Pareto). *For any two allocations $x, y \in X^N$, if*

$$U(x_i) \geq U(y_i) \quad \textit{for every } i,$$

then $x \succeq y$. If, in addition, the inequality is strict for at least one individual, then $x \succ y$.

The final core axiom is a classical equalizing-transfer principle. It requires society to weakly prefer a mean-preserving transfer that moves two individual bundles closer together.

Axiom 3.4 (Pigou-Dalton principle). *For any bundle profile $(x_1, \ldots, x_N)$ and any $i \neq j$, let*

$$x'_i = (1-\delta)x_i + \delta x_j, \qquad x'_j = \delta x_i + (1-\delta)x_j$$

with $\delta \in (0, 1/2)$, and let $x'_k = x_k$ for $k \neq i, j$. Then

$$(x'_1, \ldots, x'_N) \succsim (x_1, \ldots, x_N).$$

Together, these four axioms deliver the usual additive prioritarian form: the social criterion can be represented by a sum of concave representations of preferences.

Lemma 3.5 (Prioritarian social ordering). *A social welfare ordering satisfies Continuity, Separability, Pareto, and the Pigou-Dalton principle if and only if there exists a strictly increasing, concave and continuous function ϕ such that $\succsim$ is represented by*

$$W(x) = \sum_i \phi\left(U^{LC}(x_i)\right),$$

where U^{LC} is a least concave representation of preferences.[2] *We call such a social ordering a prioritarian social ordering.*

This lemma is a key step as it in turn provides a canonical benchmark for any cardinal measure of real income growth, through a welfare-change index $\Delta : X^N \times X^N \to \mathbb{R}$ consisting in a sum of individual indices

$$\Delta(x, y) := W(y) - W(x) = \sum_{i=1}^{N} \Big(\phi(U^{LC}(y_i)) - \phi(U^{LC}(x_i))\Big).$$

[2] A least concave representation of preferences is a concave utility function U such that for any other concave utility function $\tilde{U}$, there exists a concave function g such that $\tilde{U} = g \circ U$. Least concave representations of preferences exist whenever there is at least one concave representation of preferences (Debreu 1976), which is the case by assumption in this paper.

But lemma 3.5 is only a first step. It shows that the social ordering must take a prioritarian form, but it does not yet determine the relevant individual welfare cardinalization. The reason is that the least concave representation is not unique. More precisely, if U^{LC} is least concave, then every positive affine transform of U^{LC} is least concave as well. That indeterminacy matters because the social ranking depends on the composite curvature of $\phi \circ U^{LC}$, not on the curvature of ϕ alone. A fixed social transform ϕ can therefore encode very different effective degrees of inequality aversion depending on which least concave representation is used. The quasi-homothetic case makes this point transparent. If

$$e(p, u) = A(p) + B(p)u,$$

then for any reference price $\tilde{p}$ the money-metric utility $M_{\tilde{p}}$ is linear in income and therefore is a least concave representation:

$$M_{\tilde{p}}(p, I) = \frac{B(\tilde{p})}{B(p)} I + \left(A(\tilde{p}) - \frac{B(\tilde{p})}{B(p)} A(p) \right).$$

But the curvature of $I \mapsto \phi \circ M_{\tilde{p}}$ depends on the constant term, and that constant term varies with $\tilde{p}$. Different reference prices therefore generate different effective levels of inequality aversion even when ϕ is held fixed.[3] For this reason, the choice of the concave function ϕ and the choice of the utility representation should not be made independently.

3.3 Strengthening the fairness axiom

The proposal of the paper is to incorporate inequality aversion directly as a primitive feature of the social ordering, rather than as something added only afterwards through the choice of a concave social aggregator. Before doing so, it is useful to rewrite the Pigou-Dalton principle in income space.

Axiom 3.6 (Pigou-Dalton for income)**.** *For any common price vector p and income profile $(I_1, \ldots, I_N)$, let*

$$I'_i = I_i - \delta, \qquad I'_j = I_j + \delta,$$

with $\delta \in \big(0, (I_i - I_j)/2\big)$, and let $I'_k = I_k$ for $k \neq i, j$. Then the transformed budget profile B' is weakly preferred to the original budget profile B:

$$B' \succsim B.$$

Lemma 3.7 shows that the usual bundle-transfer principle and its income counterpart express the same minimal requirement of inequality aversion. This result comes from the fact that, for convex preferences, a direct utility function is concave for all bundles if and only if the indirect utility is concave in income for all prices (Diewert 1978).

Lemma 3.7 (Equivalence of bundle and income Pigou-Dalton principles)**.** *A prioritarian social ranking satisfies the Pigou-Dalton principle on bundles if and only if it satisfies Pigou-Dalton for income.*

[3]This is exactly the mechanism that led to the indeterminacy result in the motivating Stone-Geary example in subsection 2.3.

This equivalence makes it natural to think of inequality aversion in terms of the amount of leakage that society is willing to tolerate in an equalizing transfer of nominal income, at common prices. The difficulty, under non-homothetic preferences, is that inequality aversion cannot be summarized by a single price-independent notion of tolerated leakage in nominal-income space. The reason is that, once combined with Pareto, the social ranking must reflect the curvature of utility at the price vector under consideration, and this curvature generally varies with prices because utility cannot, in general, be represented as linear in income[4].

A good starting point is the case in which society is only as weakly inequality averse as the basic axioms force it to be. From one side, Pigou-Dalton for income requires that, at any common price vector, every *non-leaky* equalizing transfer in nominal income weakly increase social welfare. That requires summing concave representations of preferences. On the other side, in this 'maximally-utilitarian' benchmark, society does not want to tolerate any leakage beyond the minimum imposed by Pareto. This calls for summing least concave utility representations. The paper generalizes this benchmark by following the same two-step logic for any prescribed degree of inequality aversion. The first step is to decide what amount of leakage coming along an equalizing transfer should always be tolerated by the social ranking. The second step is to take the least inequality averse social ranking that always endorses such transfers. Formally, a social ranking $\succsim$ is less inequality averse than $\succsim'$ if any equal-price equalizing transfer endorsed by $\succsim$ is also endorsed by $\succsim'$. In our approach, an inequality-averse society is thus always willing to endorse leaky transfers for leakages below a certain threshold. However, it is not *a priori* willing to endorse any leaky transfers above the threshold. The axiom below formalizes this idea.

Axiom 3.8 (ϕ-prioritarianism)**.** *Let $\phi : \mathbb{R}_{++} \to \mathbb{R}$ be increasing and concave.*

Denote by $P(\phi)$ the following property. For every pair of individuals h, l such that $I_l < I_h$, for every common price vector $p \gg 0$, and for every pair (δ_l, δ_h) satisfying

$$\delta_l \geq 0, \qquad \delta_h \geq 0, \qquad \delta_l + \delta_h \leq I_h - I_l,$$

if

$$\phi(I_l + \delta_l) - \phi(I_l) + \phi(I_h - \delta_h) - \phi(I_h) \geq 0,$$

then the budget profile obtained by setting

$$I'_l = I_l + \delta_l, \qquad I'_h = I_h - \delta_h, \qquad I'_k = I_k \quad (k \notin \{l, h\})$$

is weakly preferred to the initial profile:

$$\big((p, I'_1), \ldots, (p, I'_N)\big) \succsim \big((p, I_1), \ldots, (p, I_N)\big).$$

A prioritarian social welfare ordering $\succsim$ is said to be:

- ***ϕ-admissible*** *if $\mathcal{P}(\phi)$ holds,*
- ***ϕ-prioritarian*** *if it is among the least inequality averse **ϕ-admissible** social rankings: for any other ϕ-admissible social ranking $\succsim'$, any equal-price equalizing transfer endorsed by $\succsim$ is also endorsed by $\succsim'$.*

[4] Roberts (1980) already showed that welfare prescriptions can be price-independent if and only if preferences are homothetic.

The function ϕ determines the amount of additional leakage that society is always willing to tolerate in order to achieve more equality. The more concave ϕ is, the larger the loss that can be imposed on the richer individual, relative to the gain received by the poorer one, while the transfer still counts as socially desirable. When $\phi(w) = w$, the condition

$$\phi(I_l + \delta_l) - \phi(I_l) + \phi(I_h - \delta_h) - \phi(I_h) \geq 0$$

reduces to $\delta_l \geq \delta_h$. Thus $P(\phi)$ requires the social ranking to endorse every non-leaky equalizing income transfer, which is exactly Pigou-Dalton for income.

A stronger prioritarian position enlarges the set of acceptable leaky transfers beyond the minimal benchmark, but does so in a controlled way through the choice of ϕ. Given a concave function ϕ, choosing a ranking among the least inequality averse ϕ-admissible social ranking amounts to being as close as possible to our ethical stance, without ever violating it.

The content of ϕ-prioritarianism can also be interpreted from the perspective of its consequences for growth measurement. ϕ-prioritarianism does not only discipline redistributive transfers at a point in time; it also disciplines the way income growth itself should be measured and compared across households. Suppose, for instance, that society evaluates income changes in relative terms. In that case, log-prioritarianism would imply that, at fixed prices, the real-income value of a 1% increase in nominal income be decreasing with income. A policy or time period in which prices are fixed and a poorer household undergoes a 1% decrease in nominal income while a richer household enjoys a 1% increase can thus never be socially improving.

The next subsection characterizes ϕ-prioritarian social orderings.

3.4 Characterization results

From this point onward, we impose the following regularity assumptions.

First, the common preference relation admits a representation $U : X \to \mathbb{R}_{++}$ such that, for every $p \in \mathbb{R}^L_{++}$, the map

$$u \mapsto e(p, u)$$

is twice continuously differentiable[5] on $Y := U(X)$, with

$$\frac{\partial e(p, u)}{\partial u} > 0.$$

It follows that the expenditure elasticity

$$\varepsilon(p, u) := \frac{\partial \log e(p, u)}{\partial \log u} = \frac{u e_u(p, u)}{e(p, u)}$$

is well-defined and strictly positive, and that

$$\eta(p, u) := \frac{\partial \log \varepsilon(p, u)}{\partial \log u} = \frac{u \varepsilon_u(p, u)}{\varepsilon(p, u)}$$

[5] This condition is imposed on the expenditure function, not directly on the primitive utility representation. It is therefore compatible with some nonsmooth direct representations. For instance, under Leontief preferences, Hicksian demands are linear in utility for each fixed price vector.

is well-defined and continuous in u for every p. The elasticity $\varepsilon(p, u)$ measures the percentage increase in expenditure required, at prices p, to generate a one-percent increase in utility. The term $\eta(p, u)$ records how this elasticity varies with utility.

Finally, we restrict attention to regular social rankings that admit a representation W such that, for every price profile $\mathbf{p} = (p_1, \ldots, p_N)$, the indirect social welfare function

$$(I_1, \ldots, I_N) \mapsto W\big((p_i, I_i)_{i=1}^N\big)$$

is twice continuously differentiable on $\mathbb{R}^N_{++}$.

We start with the following essential lemma:

Lemma 3.9 (Reduction to an isoelastic transfer scale)**.** *Let $\phi : \mathbb{R}_{++} \to \mathbb{R}$ be strictly increasing, concave, and C^2, with $\phi'(I) > 0$ for all $I > 0$. Let*

$$\theta_\phi := \sup_{I>0} \left\{ -\frac{I\phi''(I)}{\phi'(I)} \right\}.$$

If a regular prioritarian social ordering is ϕ-prioritarian, then $\theta_\phi < \infty$ and the ordering is also ϕ_ρ-prioritarian for

$$\rho = 1 - \theta_\phi, \qquad \phi_\rho(w) = \begin{cases} w^\rho/\rho, & \rho \neq 0, \\ \log w, & \rho = 0. \end{cases}$$

Lemma 3.9 plays a decisive role in the structure of the paper. In the definition of ϕ-prioritarianism, inequality aversion is introduced in its most general form, through an arbitrary increasing and concave transfer scale ϕ. The lemma shows that, for the purpose of identifying the least inequality-averse social rankings satisfying $P(\phi)$, this generality is not substantively relevant. What matters is not the detailed shape of ϕ over the whole income domain, but the strongest curvature restriction it imposes, namely

$$\theta_\phi := \sup_{I} \left\{ -\frac{I\phi''(I)}{\phi'(I)} \right\}.$$

Thus, although the axiom is stated for a general concave function, the admissible set of social rankings depends only on this scalar measure of curvature. The reason is closely tied to the structure of the axiom. Property $P(\phi)$ is stated for transfers in nominal income space, but it must hold at every common price vector. Because indirect utility is homogeneous of degree zero in prices and income, one can rescale all prices and incomes proportionally without changing the underlying utility situation. As a result, the axiom cannot depend on where a given level of curvature happens to be attained along the income axis: any local curvature level of ϕ can be made relevant somewhere through an appropriate rescaling of the common budget set. What therefore governs admissibility is the most restrictive curvature embodied in ϕ, that is, its maximal relative curvature. An isoelastic transfer scale with constant relative curvature equal to that value captures exactly this constraint.

This result is important because it turns the ethical content of the model into something more tractable. Instead of carrying an arbitrary concave function ϕ throughout the analysis, one can index inequality aversion by a single parameter ρ. The rest of the paper then studies how, for a given ρ, one should choose the welfare cardinalization that is least inequality-averse while remaining consistent with that ethical requirement. We first introduce a key object.

Definition 3.10 (Curvature envelope)**.** Fix $\rho \in (-\infty, 1]$. The raw curvature bound associated with ρ is

$$a^0_\rho(u) := \inf_{p\in\Delta} \{\rho\varepsilon(p, u) + \eta(p, u)\}, \qquad u \in Y.$$

The curvature envelope is the lower-semicontinuous envelope of this raw bound:

$$a_\rho(u) := \liminf_{\substack{v\to u\\ v\in Y}} a^0_\rho(v), \qquad u \in Y.$$

Thus a_ρ may take the value $-\infty$. If the raw bound a^0_ρ is already lower semicontinuous, then $a_\rho = a^0_\rho$.

We now turn to the core constructive result.

Theorem 3.11 (Existence and canonical representation)**.** *Let $\rho \in (-\infty, 1]$ and let*

$$\phi_\rho(w) = \begin{cases} \dfrac{w^\rho}{\rho}, & \text{if } \rho \neq 0,\\ \log w, & \text{if } \rho = 0. \end{cases}$$

There exists a regular ϕ_ρ-prioritarian social ordering if and only if a_ρ is finite-valued and continuous on Y. When this holds, let $u_0 \in Y$ and define

$$A_\rho(u) := \int_{u_0}^{u} a_\rho(s)\, d\log s,$$

and

$$F_\rho(u) := \int_{u_0}^{u} \exp(A_\rho(t))\, d\log t.$$

Then a social ordering is ϕ_ρ-prioritarian if and only if it is represented by

$$W(x) = \sum_i \big(F_\rho \circ U\big)(x_i).$$

Moreover, the composite representation $F_\rho \circ U$ is independent of u_0 and of the initial choice of the utility representation U (up to increasing affine transformations, innocuous for the social ranking).

To form intuition on the result, we can first look at the case where $\phi(w) = w$ ($\rho = 1$). In that case, we know that ϕ-prioritarian social orderings consist in the sum of minimally concave representations of utility. The problem is therefore to turn the arbitrary utility function U into the least concave representation, through the transformation $F_\rho \circ U$. Let $V(p, I)$ denote the indirect utility function, and let $\chi = F_\rho \circ U$. For a given price vector p, χ is concave in I, for all p if and only if:

$$\forall I, p, \qquad \chi_{II}(p, I) \leq 0 \Leftrightarrow \frac{F''(u)}{F'(u)} u \leq -\frac{V_{II}(p, I)}{V_I(p, I)^2} V(p, I)$$

where $u := V(p, I)$. But by duality results, the inequality becomes:

$$\forall p, u, \qquad \frac{F''(u)}{F'(u)} u + 1 \leq \epsilon(p, u) + \eta(p, u)$$

Finally, in order for χ to be the least concave representation of utility, this inequality must be an equality for at least one price (or at the limit of a sequence of prices), hence

$$\frac{F''(u)}{F'(u)} u + 1 = \inf_{p \in \Delta} \{\epsilon(p, u) + \eta(p, u)\}$$

Integrating this equality yields the formula given by the theorem[6]. The general case for any isoelastic ϕ can be obtained immediately by noting that the property $\mathcal{P}(\phi)$ is equivalent to the statement that each f_p must be concave on $\phi((0, \infty))$, where

$$\chi_p(I) := F(V(p, I)), \qquad f_p := \chi_p \circ \phi^{-1}.$$

The lower envelope $a_\rho(u)$ identifies the least amount of local curvature that the demand system leaves available to the social evaluator at each utility level. Integrating that envelope produces the correcting transform F_ρ.

A key consequence of the ϕ-prioritarian approach is that the relevant canonical transform $F_\rho \circ U$ depends on the level of inequality aversion. The intuition is that the function for which concavity matters depends on the value of ρ. For instance, if $\rho = 1$, then $F_\rho \circ U$ itself should be such that it is (least) concave. But whenever ρ is strictly lower than 1, concavity of the transform is not the relevant criterion anymore, since social welfare functions are not directly summing $F_\rho \circ U$ levels anymore - they are summing concave transformations of these.

From the social welfare construction of 3.11, it is possible to derive a natural canonical concept of real income through the definition below.

Definition 3.12 (Canonical real income)**.** Let $\rho \in (-\infty, 1]$, assume there exists a ϕ_ρ-prioritarian social ordering, and let F_ρ be the canonical transform constructed in Theorem 3.11. A *(canonical) real-income representation* is any utility representation $U^* : X \to \mathbb{R}_{++}$ such that

$$\phi_\rho \circ U^* \sim F_\rho \circ U,$$

where $\sim$ denotes equality up to a positive affine transformation. Whenever such a representation exists, the social ranking can be written in the usual prioritarian form

$$W(x) = \sum_i \phi_\rho\big(U^*(x_i)\big).$$

The following corollary shows that this definition coincides with money-metric utility whenever preferences are homothetic.

[6]When $\phi(w) = w$, this result recalls Theorem 5.6 in Kannai (1977), which addresses the construction of least concave utility functions - though Kannai's approach is considerably more involved. Appendix B.10 establishes that the two constructions are in fact equivalent.

Corollary 3.13 (The homothetic case). *Let $\rho \in (-\infty, 1]$, and assume preferences are homothetic. Then, for any reference price $\tilde{p}$, the money-metric utility $M_{\tilde{p}}$ is a canonical real-income representation.*

Indeed, under homothetic preferences, one can choose a homogeneous utility representation U which satisfies $\varepsilon(p, u) \equiv 1$ and $\eta(p, u) \equiv 0$. This implies $a_\rho(u) \equiv \rho$, and hence the canonical transform satisfies

$$F_\rho(u) \sim \phi_\rho(u).$$

Since U is homogeneous, any money-metric utility $M_{\tilde{p}}$ is a positive affine transform of U. Hence

$$(\phi_\rho \circ M_{\tilde{p}}) \sim (\phi_\rho \circ U) \sim (F_\rho \circ U),$$

so $M_{\tilde{p}}$ is a canonical real-income representation. For homothetic preferences, ϕ-prioritarianism therefore prescribes the use of money-metric utility functions as arguments of prioritarian social welfare functions.

The next theorem generalizes the link between canonical real income and money-metric utility functions for non-homothetic preferences. We start with a generalized definition of money-metric utility, slightly broader than the standard one.

Definition 3.14 (Extended money-metric). Let $\tilde{p} = (p_n)_{n\in\mathbb{N}}$ be a sequence of price vectors such that for every u,

$$\bar{\varepsilon}(u) := \lim_{n\to\infty} \varepsilon(p_n, u) \in \mathbb{R}_{++}$$

and $\bar{\varepsilon}$ is continuous. Fix $U_0 > 0$. The extended money-metric utility associated with $\tilde{p}$ is

$$M_{\tilde{p}}(x) := \exp \int_{U_0}^{U(x)} \bar{\varepsilon}(u) d \log u$$

Definition 3.14 enlarges the usual money-metric construction by allowing the reference price to be approached along a sequence. When $\tilde{p}$ is the constant sequence equal to a fixed price, $M_{\tilde{p}}$ collapses to the usual money-metric (up to a scaling constant). Otherwise, $M_{\tilde{p}}$ is still a monotone transformation of U and is therefore a valid representation of preferences. This extension will turn out useful in some cases of interest here, when the canonical welfare representation selected by ϕ_ρ-prioritarianism corresponds to a boundary price system rather than to a single finite interior reference price. We can now characterize when money-metric is a valid representation of canonical real income.

Theorem 3.15 (When money-metric is a representation of real income). *Fix $\rho \in (-\infty, 1]$ and assume that a ϕ_ρ-prioritarian social ordering exists. Let $\tilde{p} = (p_n)_{n\in\mathbb{N}} \subset \Delta$ be a sequence of price vectors such that, for every $u > 0$,*

$$\bar{\varepsilon}(u) := \lim_{n\to\infty} \varepsilon(p_n, u) \in \mathbb{R}_{++}$$

Then the extended money-metric utility $M_{\tilde{p}}$ is a representation of real income if and only if

1. *$\bar{\varepsilon}(u)$ is continuously differentiable,*

2. $\forall u > 0$,

$$\rho\,\bar{\varepsilon}(u) + \bar{\eta}(u) = a_\rho(u), \qquad \bar{\eta}(u) := \frac{\partial \log \bar{\varepsilon}(u)}{\partial \log u}$$

A sufficient condition for (2) - given condition (1) - is that

$$\tilde{p} \in \bigcap_{u>0} \arg\inf_{p\in\Delta} \{\rho\,\varepsilon(p,u) + \eta(p,u)\},$$

together with the assumption that $\bar{\eta}(u) = \lim_n \frac{\partial \log \varepsilon(\tilde{p}_n,u)}{\partial \log u}$.

If such a price $\tilde{p}$ exists, then, for every utility level u, the same reference price system $\tilde{p}$ is the one that minimizes the local curvature term $\rho\varepsilon(p,u) + \eta(p,u)$[7]. Hence the lower envelope is not merely an abstract infimum over prices, but is actually traced out by a single price system:

$$a_\rho(u) = \rho\bar{\varepsilon}(u) + \bar{\eta}(u).$$

Canonical real income can therefore be expressed as the extended money-metric utility associated with $\tilde{p}$. In that case, the normalization selected by ϕ_ρ-prioritarianism admits a simple economic interpretation: real income is obtained by evaluating utility at a common reference price system, possibly reached only in the limit along a sequence of prices.

What remains unclear is the interpretation of the corresponding reference price $\tilde{p}$. The following corollary shows that after adding some mild structure on preferences, the economic intuition behind the selection of the reference price system greatly simplifies for inequality aversion at least as strong as log.

Corollary 3.16 (A sufficient condition for a money-metric representation when $\rho \leq 0$)**.** *Fix* $\rho \in (-\infty, 0]$ *and assume that a* ϕ_ρ*-prioritarian social ordering exists. Let* $\tilde{p} = (p_n)_{n\in\mathbb{N}} \subset \Delta$ *be a sequence of price vectors such that, for every* $u > 0$,

$$\bar{\varepsilon}(u) := \lim_{n\to\infty} \varepsilon(p_n, u) \in \mathbb{R}_{++}$$

Assume

(Efficiency dynamics). *For every* $u > 0$,

$$\arg\sup_{p\in\Delta} \varepsilon(p,u) \subset \arg\inf_{p\in\Delta} \eta(p,u).$$

[7]When an optimizer is not attained in the open simplex Δ, expressions such as

$$\tilde{p} \in \arg\inf_{p\in\Delta} h(p,u) \quad \text{or} \quad \tilde{p} \in \arg\sup_{p\in\Delta} h(p,u)$$

are understood asymptotically: $\tilde{p} = (p_n)_n \subset \Delta$, and

$$h(p_n,u) \to \inf_{p\in\Delta} h(p,u) \quad \text{or} \quad h(p_n,u) \to \sup_{p\in\Delta} h(p,u),$$

respectively.

Then a sufficient condition to replace condition (2) in 3.15 is that

$$\tilde{p} \in \bigcap_{u>0} \arg\sup_{p\in\Delta}\{\varepsilon(p,u)\},$$

together with the assumption that $\bar{\eta}(u) = \lim_n \frac{\partial \log \varepsilon(\tilde{p}_n, u)}{\partial \log u}$.

Finally, let $s \mapsto x_s$ *be a* $\mathcal{C}^1$ *optimal consumption path associated with a* $\mathcal{C}^1$ *continuous budget path* $s \mapsto (p_s, I_s)$*. Writing* $u_s := U(x_s)$*, real income satisfies*

$$U^*(x_t) = U^*(x_0)\, \exp\left(\int_0^t \frac{\sup_{\tilde{p}\in\Delta} \varepsilon(\tilde{p}, u_s)}{\varepsilon(p_s, u_s)}\, \mathrm{Div}(s)\, ds\right),$$

where

$$\mathrm{Div}(s) := \frac{p_s \dot{x}_s}{p_s x_s}$$

is the instantaneous quantity Divisia index.

This inclusion condition,

$$\arg\sup_{p\in\Delta} \varepsilon(p,u) \subset \arg\inf_{p\in\Delta} \eta(p,u),$$

is an ordinal restriction on the type of non-homotheticity under study. For each price system p, the utility-expenditure elasticity $\varepsilon(p,u)$ measures the income increase required to reach the next utility level; equivalently, $1/\varepsilon(p,u)$ is the local efficiency of income in producing utility.[8] The term $\eta(p,u)$ measures how this efficiency changes with utility. A low value of $\eta(p,u)$ means that income efficiency deteriorates slowly as utility rises. The condition therefore requires that, whenever income is already least efficient at a given indifference curve, it is also precisely there that its efficiency declines the slowest along that curve.

This can be related to the mechanism behind non-homotheticity. Suppose necessity goods are more efficient than luxury goods, in the ordinal sense that low values of $\varepsilon(p,u)$ are associated with price systems under which the Hicksian budget shares are more concentrated on necessity goods. Since the budget share of necessity goods decreases with utility, efficiency at such prices also tends to fall as utility rises. By contrast, if one considers prices at which income is already least efficient at a given utility level, expenditure is by assumption already tilted toward luxury goods, which are by definition the goods that dominate at higher utility levels. The compositional effect of further increases in utility is then weaker, so efficiency declines more slowly. This is the intuition behind the inclusion condition.

[8]Indeed,

$$\frac{1}{\varepsilon(p,u)} = \frac{\partial \log V(p,I)}{\partial \log I} \qquad \text{at } I = e(p,u).$$

Example. A useful benchmark is the Gorman polar form. When

$$e(p,u) = A(p) + C(p)u,$$

one has

$$\varepsilon(p,u) = \frac{u\,C(p)}{A(p) + u\,C(p)}, \qquad \eta(p,u) = \frac{A(p)}{A(p) + u\,C(p)} = 1 - \varepsilon(p,u).$$

Hence, for every utility level u, maximizing $\varepsilon(p,u)$ is exactly equivalent to minimizing $\eta(p,u)$, so the inclusion condition holds trivially. The economic intuition is also transparent. In the Gorman form, $A(p)$ is the expenditure needed to cover the subsistence component of consumption, while $C(p)$ is the marginal cost of increasing utility beyond that baseline. Prices at which income is least efficient are therefore those that minimize the ratio $A(p)/C(p)$, that is, those at which subsistence expenditure is smallest relative to discretionary expenditure. At such prices, the expenditure pattern is already as close as possible to the high-utility composition, so further increases in utility generate no additional compositional effect on efficiency, and hence it declines the slowest.

To build further intuition for the characterization results above, the next section works through three standard non-homothetic demand systems - Gorman polar form, PIGL specifications, and non-homothetic CES - and derives the corresponding canonical real-income formulas.

4 Analytic examples under standard specifications

This section applies the characterization results to three standard classes of non-homothetic preferences: Gorman polar form, PIGL preferences, and non-homothetic CES. Each example answers the same three questions. First, what real-income representation is selected by ϕ_ρ-prioritarianism? Second, when can that representation be interpreted as a money-metric utility at an endogenous reference price system? Third, how does the resulting real-income growth rate relate to a Divisia quantity index?

Throughout the section, inequality aversion is assumed to be at least as strong as log, so that $\rho \leq 0$. This restriction keeps the main text focused on the empirically most relevant case and gives the selected reference prices a simple interpretation. The calculations behind the formulas all follow the same logic. For each demand system, the appendix computes the local curvature term selected by Theorem 3.11, identifies the price system that minimizes it, and then translates the resulting welfare scale into a real-income representation. The main text reports only the economic content of this exercise. The appendix reports the corresponding calculations (including for $\rho > 0$). The notation $\sim$ denotes equality of utility representations up to the innocuous normalizations allowed by Theorem 3.11.

The common lesson is that the canonical real-income scale is generally close to a familiar utility or money-metric object, but only after the reference system has been selected by the curvature-minimization problem in Section 3. Arbitrary initial-price or final-price money-metrics therefore need not coincide with the welfare scale implied by the chosen degree of prioritarian concern.

4.1 Gorman polar form

The Gorman polar form is the benchmark case in which non-homotheticity enters through an additive subsistence term. Suppose that the expenditure function is

$$e(p,u) = A(p) + C(p)u,$$

where $A(p)$ and $C(p)$ are homogeneous of degree 1, and let U denote the underlying utility representation. In this specification, the curvature-minimization problem has a simple interpretation: it selects the price system at which the subsistence component is cheapest relative to the marginal cost of utility. This is why the ratio $A(p)/C(p)$ plays the central role below.

Canonical representation. Let

$$c := \inf_p \frac{A(p)}{C(p)}.$$

A canonical real-income representation is given by

$$U^*(x) \sim U(x) + c.$$

The key insight is that, within the Gorman form framework, measuring welfare via a money-metric transformation with arbitrary reference prices is misguided: the original utility is already close to the appropriate normalization, and the correction amounts to a mere (yet non-innocuous) additive constant.

Money-metric interpretation. A canonical real-income representation is the extended money-metric utility associated with any reference price $\tilde{p}$ such that

$$\tilde{p} \in \arg\inf_p \frac{A(p)}{C(p)}.$$

The Gorman form thus provides a polar case in which the reference price selected by ϕ_ρ-prioritarianism admits a particularly simple interpretation: it is the price system that minimizes subsistence expenditure $A(p)$ relative to discretionary expenditure $C(p)$.

Growth formula. Assume $c = 0$ and consider a $\mathcal{C}^1$ path $s \mapsto (x_s, p_s, I_s)$ between dates 0 and t. The real-income path can be written, up to the usual innocuous normalization, as

$$U^*(x_t) \sim \exp \int_0^t \Lambda(s)\, \mathrm{Div}(s)\, ds, \qquad \Lambda(s) = \frac{I_s}{I_s - A(p_s)}.$$

The Divisia index therefore receives a simple correction: the larger the share of expenditure devoted to subsistence needs ($A(p_s)$), the more Divisia quantity growth translates into welfare growth. $\Lambda(s)$ converges to 1 as income levels rise, and the canonical real-income index converges to a homothetic index.[9]

[9] In order to make the section more homogeneous, we assumed that $\rho \leq 0$ throughout, but in the case of the Gorman form, it is easy to check that the results are still true for positive values of ρ.

Example (Stone-Geary). For Stone-Geary preferences

$$U(x_1, \ldots, x_L) = \prod_i (x_i - \gamma_i)^{\beta_i}, \qquad \sum_i \beta_i = 1,$$

one obtains

$$c = \prod_i \gamma_i^{\beta_i} \qquad \Longrightarrow \qquad U^*(x) \sim U(x) + \prod_i \gamma_i^{\beta_i}.$$

Let $\tilde{x}_i = x_i - \gamma_i$ denote discretionary consumption and

$$\tilde{s}_i = \frac{p_i \tilde{x}_i}{\sum_j p_j \tilde{x}_j}$$

the corresponding discretionary budget shares. When at least one of the consumption goods is non-essential, then $\prod_i \gamma_i^{\beta_i} = 0$, and

$$U^*(x_t) \sim \exp \int_0^t \sum_i \tilde{s}_i \frac{\tilde{x}_i'(s)}{\tilde{x}_i(s)}\, ds.$$

Figure 5 plots the log-prioritarian social growth rate from the third motivating example of Section 2. The log-prioritarian approach thus settles the indeterminacy: in this stylized environment, the policy leads to a strong decrease in social welfare and should therefore not be approved.

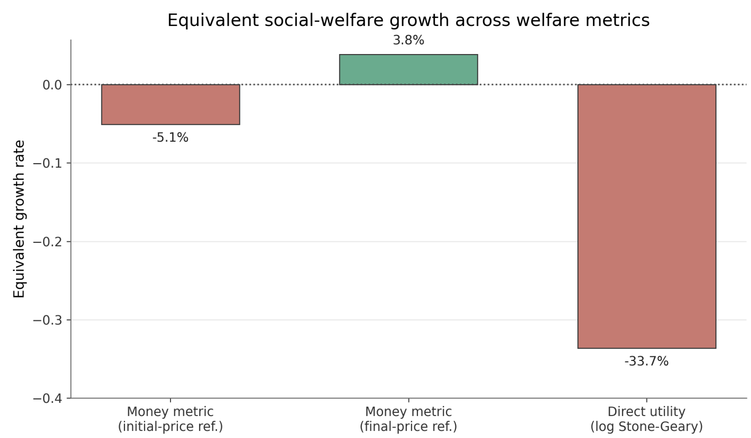

Figure 5: Growth of the log-prioritarian social welfare function

4.2 PIGL

The PIGL specification, introduced by Muellbauer (1975), extends the Gorman form by allowing for non-linear Engel curves. The expenditure function is

$$e(p, u) = \left[(1 - u) f(p)^\kappa + u g(p)^\kappa\right]^{1/\kappa},$$

where $\kappa > 0$ and both $f(p)$ and $g(p)$ are homogeneous of degree 1. Let U denote the underlying utility function, and define

$$A(p) = f(p)^\kappa, \qquad B(p) = g(p)^\kappa, \qquad C(p) = B(p) - A(p).$$

The PIGL case preserves the same structure as the Gorman polar form, but with a nonlinear transformation of the utility scale. As shown in the Appendix, for $\rho \leq 0$ the curvature-minimizing reference system is again obtained by minimizing the ratio $A(p)/C(p)$.

Canonical representation. Once again, let

$$c := \inf_p \frac{A(p)}{C(p)}.$$

A canonical real-income representation is given by

$$U^*(x) \sim (U(x) + c)^{1/\kappa},$$

which naturally generalizes the Gorman form.

Money-metric interpretation. The above canonical real-income representation corresponds to the extended money-metric utility associated with any reference price $\tilde{p}$ such that

$$\tilde{p} \in \arg\inf_p \frac{A(p)}{C(p)}.$$

As before, these reference prices minimize subsistence expenditure $A(p)$ relative to discretionary expenditure $C(p)$.

Growth formula. Assume $c = 0$ and consider a $\mathcal{C}^1$ path $s \mapsto (x_s, p_s, I_s)$ between dates 0 and t. The canonical real-income path can be written, up to the usual multiplicative normalization, as

$$U^*(x_t) \sim \exp \int_0^t \Lambda(s)\, \mathrm{Div}(s)\, ds, \qquad \Lambda(s) = \frac{I_s^\kappa}{I_s^\kappa - f(p_s)^\kappa}.$$

The term $\frac{I_s^\kappa}{I_s^\kappa - f(p_s)^\kappa}$ is strictly decreasing in κ, which means that the non-homotheticity correction is larger for lower values of κ. The intuition comes from writing the indirect utility function associated with PIGL:

$$V(p, I) = \frac{I^\kappa - f(p)^\kappa}{g(p)^\kappa - f(p)^\kappa} = \frac{\left(\frac{I}{f(p)}\right)^\kappa - 1}{\left(\frac{g(p)}{f(p)}\right)^\kappa - 1} \sim \left(\frac{I}{g(p)}\right)^\kappa,$$

since $\frac{I}{f(p)} > 1$ and $\frac{g(p)}{f(p)} > 1$. In words, $V(p, I)$ is closer to representing homothetic preferences when κ is large.

Example. Consider a subclass of PIGL commonly used in the applied literature studying income effects, given as in Boppart (2014) by:

$$V(p, I) = \frac{1}{\kappa}\left[\left(\frac{I}{B(p)}\right)^\kappa - 1\right] - \frac{\nu}{\gamma}\left[\left(\frac{D(p)}{B(p)}\right)^\gamma - 1\right],$$

where $\kappa, \gamma \in (0,1)$ and $\nu > 0$.

Keeping our previous notations, one finds

$$\inf_p \frac{A(p)}{C(p)} = \frac{1}{\kappa} - \frac{\nu}{\gamma}.$$

A canonical real-income representation is therefore equal, up to the usual innocuous normalization, to

$$V^*(p,I) \sim \left[\frac{1}{\kappa}\left(\frac{I}{B(p)}\right)^{\kappa} - \frac{\nu}{\gamma}\left(\frac{D(p)}{B(p)}\right)^{\gamma}\right]^{1/\kappa}.$$

Once again, under ϕ-prioritarianism, the appropriate approach is to work with this slightly modified version of the original utility - not a money-metric at an arbitrary reference point.

4.3 Non-homothetic CES

Finally, we consider the non-homothetic CES specification introduced by Hanoch (1975) and used by Comin, Lashkari, and Mestieri (2021) to measure income effects. It is defined implicitly by

$$\sum_i \Omega_i^{1/\sigma}\left(\frac{x_i}{U^{\epsilon_i}}\right)^{\frac{\sigma-1}{\sigma}} = 1,$$

where x_i is the consumption of good i, σ is the constant elasticity of substitution, and ϵ_i is a utility-elasticity parameter for good i. We assume $\epsilon_1 \leq \cdots \leq \epsilon_L$, and that goods are gross complements ($\sigma < 1$), following the literature that uses nh-CES specifications to model non-homothetic consumption. This implies that good L is the most luxurious good. The case $\sigma > 1$ is treated in the appendix.

Let

$$w_i(p,u) = \frac{\Omega_i p_i^{1-\sigma} u^{(1-\sigma)\epsilon_i}}{\sum_j \Omega_j p_j^{1-\sigma} u^{(1-\sigma)\epsilon_j}}$$

denote Hicksian budget shares.

Canonical representation. For every value of $\rho \leq 0$, a canonical real-income representation is given by

$$U^*(x) \sim U(x)^{\epsilon_L}.$$

In particular, when $\epsilon_L = 1$, the ethically selected canonical real-income index is simply the original utility function. For every value of $\rho \leq 0$, ϕ_ρ-prioritarianism therefore selects the original utility function, once the utility-elasticity parameters ϵ are normalized in the right way, namely $\epsilon_L = 1$.[10]

[10] The normalization of the ϵ_i is actually innocuous when $\rho = 0$, since $\log U^a = a \log U$, and the two thus only differ by a harmless multiplicative factor.

Money-metric interpretation. The corresponding reference prices are boundary prices:

- if $\rho < 0$, the extended money-metric puts all weight on the most luxurious good: $p_L = 1$ and $p_i = 0$ for $i \neq L$;
- if $\rho = 0$, any boundary price concentrating weight on a single good is admissible.

The reference system is pushed to the boundary so as to make all goods free relative to the cost of the most luxurious good; the interpretation is thus the same as for the PIGL and Gorman form results. The growth formula gives a directly observable interpretation of this result. Instead of depending on the unobserved utility-elasticity parameters alone, the correction can be written in terms of the dispersion of Marshallian income elasticities along the income distribution.

Growth formula. Consider a C^1 Marshallian path $s \mapsto (p_s, I_s, x_s)$ and write $u_s = U(x_s)$. Then

$$U^*(x_t) = U^*(x_0) \exp\left(\int_0^t \Lambda(p_s, I_s)\,\mathrm{Div}(s)\,ds\right),$$

where

$$\mathrm{Div}(s) = \frac{p_s \dot{x}_s}{p_s x_s}$$

and

$$\Lambda(p, I) = \exp\left[\frac{1}{1-\sigma}\int_I^\infty \mathrm{Var}_{w(p,z)}\big(\epsilon^I(p,z)\big)\,d\log z\right].$$

Here

$$\epsilon_i^I(p,z) := \frac{\partial \log x_i(p,z)}{\partial \log z}, \qquad \mathrm{Var}_{w(p,z)}(\epsilon^I) = \sum_i w_i(p,z)\big(\epsilon_i^I(p,z)\big)^2 - 1.$$

Canonical real-income growth is a Divisia quantity-growth term multiplied by a non-homotheticity correction Λ. The correction is entirely driven by the elasticity of substitution and the budget-share-weighted dispersion of income elasticities across goods. Those elasticities can be backed out from how budget shares move along the income distribution. In the homothetic benchmark, all those elasticities coincide and the variance term vanishes, so $\Lambda \equiv 1$.

Illustrative calibration with US expenditure data To illustrate in a 'toy calibration' how this semi-parametric growth formula can be used, we use the Bureau of Labor Statistics Consumer Expenditure Survey, which reports mean annual expenditures on goods and services by before-tax income decile at the consumer unit level[11]. Because the average number of individuals per consumer unit varies across deciles, we re-normalize all expenditures by the square root of the number of individuals in the unit. This has no effect on the computation of

[11] According to the glossary of the Bureau of Labor Statistics, "consumer units consist of families, single persons living alone or sharing a household with others but who are financially independent, or two or more persons living together who share major expenses".

budget shares but could influence the computation of the expenditure elasticities of demand[12]. Let E_{id} denote (equivalized) expenditure of decile d on category i.

We work at the most aggregated level so that the retained categories can be plausibly considered as gross complements, as assumed in the theoretical derivations. We exclude 'Personal insurance and pensions' in order to stay in the realm of consumption goods and services, as well as 'Education' expenditure, whose variation across deciles is very sensitive to household composition effects. We end up with twelve non-overlapping groups: Food, Alcoholic beverages, Housing, Apparel and services, Transportation, Healthcare, Entertainment, Personal care products and services, Reading, Tobacco products and smoking supplies, Miscellaneous, and Cash contributions. Budget shares are re-computed using the total expenditure on retained categories, $w_{id} = E_{id}/\sum_j E_{jd}$. Figure 6 shows log budget shares as a function of log expenditure. In theory, the expenditure elasticities of demand $\varepsilon(p, z)$ depend on the expenditure level z. One could therefore locally estimate the elasticities by computing the local derivatives of Engel curves with a discrete approximation. However, the irregularities in the curves, potentially reflecting data noise rather than actual consumer behavior, mechanically induce a high variance in expenditure elasticities. We instead choose a conservative approach by approximating the Engel curves with a linear interpolation between the first decile and the last decile. Considering the monotonic shape of each Engel curve, this approximation correctly captures the first-order direction of the Engel gradients, as shown in Figure 6.

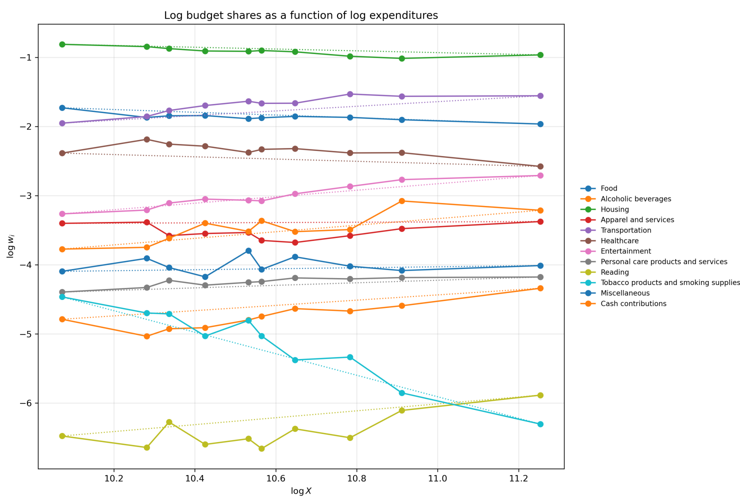

Figure 6: Log budget shares as a function of log expenditures

This approach implies a constant ε_i, estimated with:

$$\tilde{\varepsilon}_i := 1 + \frac{\log w_{i,10} - \log w_{i,1}}{\log X_{10} - \log X_1}.$$

Figure 7 plots these estimated elasticities.

[12]The appendix presents results when expenditures are not equivalized - the differences are small.

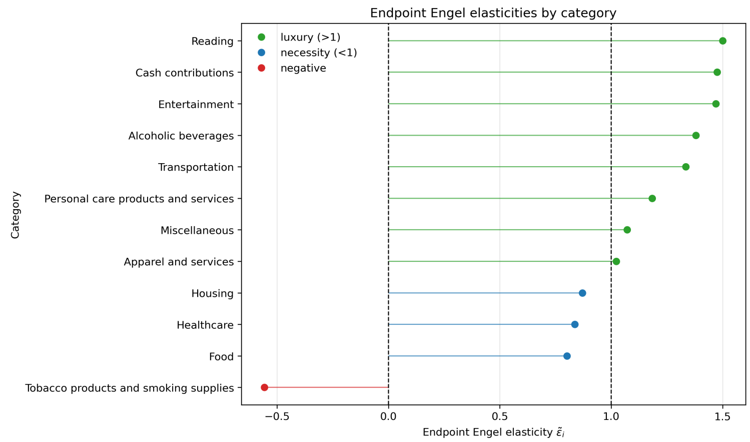

Figure 7: Engel elasticities by category

Using these endpoint elasticities together with decile-specific expenditure shares, we then compute the weighted cross-good dispersion

$$\mathrm{Var}_{w_d}(\tilde{\varepsilon}) = \sum_i w_{id} \left(\tilde{\varepsilon}_i - \sum_j w_{jd} \tilde{\varepsilon}_j \right)^2 .$$

Finally, setting $\sigma = 0.26$ as in Comin, Lashkari, and Mestieri (2021), we approximate the nh-CES correction by computing the area below $\mathrm{Var}_{w_x}(\tilde{\varepsilon})$, before scaling it by $\frac{1}{1-\sigma}$.

$$\widehat{\Lambda}_d = \exp\left(\frac{1}{1-\sigma} \int_{X_d}^{X_{10}} \mathrm{Var}_{w_x}(\tilde{\varepsilon})\, d\log x \right) .$$

The resulting correction, plotted in Figure 8, is monotone and modest: it declines from about 1.08 in the first decile to 1.0 in the tenth (which holds by construction). Therefore, in this illustrative calibration, relative to the top decile normalization, the correction for the first decile is about 8%.

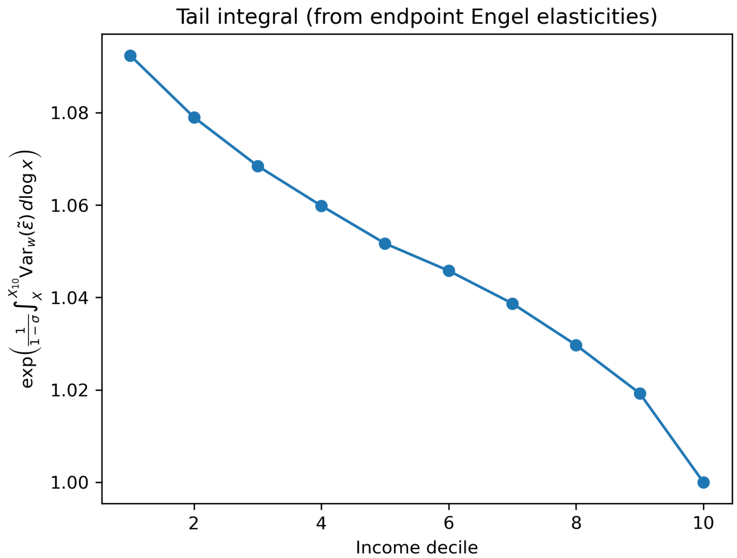

Figure 8: Non homotheticity correction from a nh-CES specification ($\sigma = 0.26$)

It is instructive to compare this magnitude to a discretionary income approach that would be induced by a Gorman form specification. Following the poverty threshold used by the US Census Bureau, we set a subsistence expenditure equal to 15,000 and plot the induced non-homotheticity correction in Figure 9. The correction ranges from 2.7 in the first decile to 1.2 in the tenth, which is much larger than the nh-CES correction. This comparison shows that quantitative welfare corrections may depend massively on the demand structure one imposes.

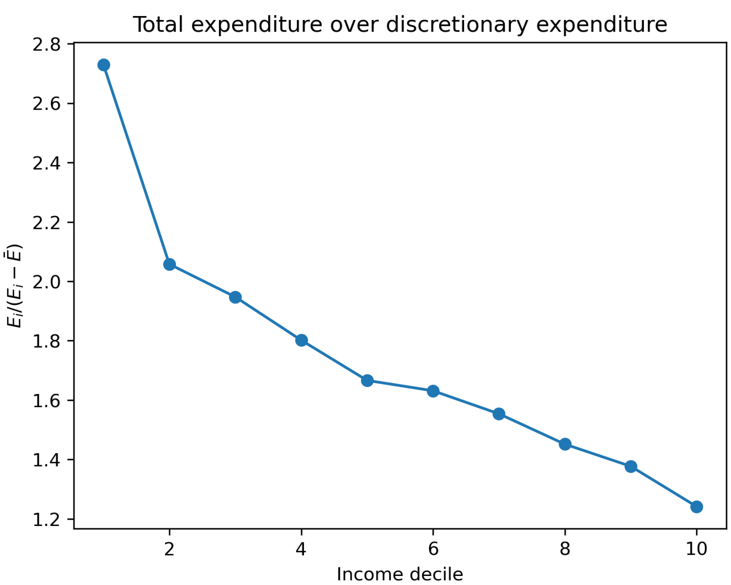

Figure 9: Non homotheticity correction with Gorman form

This section has thus shown that ϕ-prioritarianism yields tractable results amenable to empirical work on income effects. However, following the spirit of section 3, we deliberately fixed a common preference map. That restriction allowed the paper to isolate one problem cleanly: how the choice of welfare cardinalization interacts with finite inequality aversion once interpersonal comparability is temporarily set aside. But it also leaves out a distinct source of measurement and welfare ambiguity, namely heterogeneity in preferences across households or over time. The next section extends ϕ-prioritarianism to a framework with heterogeneous preferences.

5 Extension to heterogeneous preferences

This section asks what survives once preferences are allowed to differ across individuals, and shows that the logic of ϕ-prioritarianism can still be applied, both in a *unique-profile* and *multi-profile* framework. In the unique-profile framework, the social evaluator ranks allocations for one fixed collection of preference relations. In the multi-profile framework, the evaluator specifies a ranking for each admissible preference profile, using a common rule across profiles.

5.1 Unique preference profile framework

Assume there is a population of N consumers with heterogeneous preference[13]s

$$\mathcal{R}_N = (R_1, ..., R_N) \in \mathcal{R}^N.$$

We are now interested in social rankings consisting of weak orders for that specific preference profile. We will focus on a specific class of social rankings, which we call $\mathcal{U}$-utilitarian social rankings. For a bundle x and preference relation R, $I(x, R)$ denotes the indifference set belonging to the indifference map induced by R and passing through x, and $\mathcal{I}_N = \{I(x, R), x \in X, R \in \mathcal{R}_N\}$. We say that an indifference set $I \in \mathcal{I}_N$ is *above* $I' \in \mathcal{I}_N$ if, for every $x' \in I'$, there exists $x \in I$ such that $x \gg x'$. A function $\mathcal{U} : \mathcal{I}_N \to \mathbb{R}$ is said to be increasing if

$$\mathcal{U}(I) > \mathcal{U}(I')$$

whenever I is above I'

Definition 5.1. Let $Y \subseteq \mathbb{R}_{++}$ be an interval and let $\mathcal{U} : \mathcal{I}_N \to Y \subseteq \mathbb{R}_{++}$ be an increasing function such that, for every $R \in \mathcal{R}_N$,

$$U_R^{\mathcal{U}}(x) := \mathcal{U}(I(x, R)) \quad \text{has image} \quad U_R^{\mathcal{U}}(X) = Y.$$

A social ordering is said to be $\mathcal{U}$-utilitarian if it admits a representation of the form

$$W(x) = \sum_{i=1}^{N} f\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

for some increasing function $f : Y \to \mathbb{R}$. W is *regular* if, for every price profile, the induced indirect welfare function is twice continuously differentiable in income.

[13] We assume each preference is shared by at least two consumers (so that fairness principles among equal preferences have bite).

For our purposes, the essential feature of $\mathcal{U}$-utilitarian rankings is that they aggregate individual well-being indices additively. The common-image restriction mainly simplifies the representation theorem.[14] The fact that these indices depend only on the indifference set reached by each individual is also not essential to the logic of ϕ-prioritarianism, but it is consistent with standard contour-independence requirements, such as Hansson independence, according to which social comparisons should depend only on the individual indifference curves passing through the bundles being compared.[15]

The class also includes the opportunity-equivalent welfare criteria studied by Piacquadio (2017). These criteria take the form

$$x \mapsto \sum_i \phi \circ G \circ U_i(x_i), \tag{2}$$

where the U_i are opportunity-equivalent well-being functions, G is a joint concavification, and ϕ adds further inequality aversion.[16] Since opportunity-equivalent well-being functions depend only on indifference sets, they are special cases of $\mathcal{U}$-utilitarian rankings. The present framework is more general because the contour scale $\mathcal{U}$ need not itself be generated by an opportunity-equivalent construction.

For the rest of the paper, we fix a contour scale $\mathcal{U}$. On the class of $\mathcal{U}$-utilitarian social rankings, ϕ-prioritarianism extends very easily to the heterogeneous-preference setting, by weakening it such that it only applies to individuals sharing the same preferences.[17]

Axiom 5.2 (ϕ-prioritarianism among equals). *Let $\phi : \mathbb{R}_{++} \to \mathbb{R}$ be increasing and concave. Denote by $P(\phi)$ the following property. For every pair of individuals h, l such that $m_l < m_h$ and $R_h = R_l$, for every common price vector $p \gg 0$, and for every pair (δ_l, δ_h) satisfying*

$$\delta_l \geq 0, \qquad \delta_h \geq 0, \qquad \delta_l + \delta_h \leq m_h - m_l,$$

if

$$\phi(m_l + \delta_l) - \phi(m_l) + \phi(m_h - \delta_h) - \phi(m_h) \geq 0,$$

[14] Such a property can be imposed by an interpersonal axiom which Piacquadio (2017) calls 'Possibility of trade-offs', and that requires that any well-being level attainable by a given individual can be attained by any other individual.

[15] One may view Hansson independence as a weakening of Arrow's independence of irrelevant alternatives. Together with Pareto, it also captures a form of "preference monotonicity" (Fleurbaey and Maniquet 2017; Maskin 1978). Suppose, for example, that two individuals A and B consume the same bundle, but that this bundle occupies a higher position in B's preference ordering than in A's: every bundle that was below it for A remains below it for B, while some bundles that were above it for A now lie below it for B. Then it is natural to regard individual B as at least as well off as individual A. $\mathcal{U}$-utilitarian rankings are aligned with this idea, because the example above corresponds exactly to looking at whether one indifference set is above another.

[16] Opportunity sets (Thomson 1994) are a collection of consumption sets $(C_\lambda)_{\lambda \geq 0}$ that are nested in the sense that $\lambda' \geq \lambda \Rightarrow C_\lambda \subset C_{\lambda'}$. An individual's situation is evaluated by taking the lowest λ such that they are indifferent between their actual bundle and their favorite bundle in C_λ. This class is prominent in the literature (Bosmans, Decancq, and Ooghe 2018), with its most well-known instances being the money-metric and ray utility functions.

[17] This idea in a framework with heterogeneous preferences was introduced by Fleurbaey (2005) for weakening the Pigou-Dalton principle.

then the budget profile obtained by setting

$$m'_l = m_l + \delta_l, \qquad m'_h = m_h - \delta_h, \qquad m'_k = m_k \quad (k \notin \{l, h\})$$

is weakly preferred to the initial profile:

$$\big((p, m'_1, R_1), \dots, (p, m'_N, R_N)\big) \succsim \big((p, m_1, R_1), \dots, (p, m_N, R_N)\big).$$

A regular $\mathcal{U}$-utilitarian social welfare ordering $\succsim$ is said to be:

- ϕ***-admissible*** *if $P(\phi)$ holds,*
- ϕ***-prioritarian*** *if it is among the least inequality averse regular* ϕ***-admissible*** *and $\mathcal{U}$-utilitarian social rankings: for any other regular ϕ-admissible $\mathcal{U}$-utilitarian social ranking $\succsim'$, any equal-price and equal-preference equalizing transfer endorsed by $\succsim$ is also endorsed by $\succsim'$.*

We now move on to the characterization result.

5.2 Characterization result

We can now state the heterogeneous-preference analogue of the characterization result. The only additional step is to compute the relevant curvature bound after pooling across the preference relations present in the fixed profile.

Given the fixed contour scale $\mathcal{U}$, define

$$e^{\mathcal{U}}(p, u, R) := \min\{px : U_R^{\mathcal{U}}(x) \geq u\}.$$

Similarly to the previous section, we assume that $\mathcal{U}$ is regular in the following sense. For every $R \in \mathcal{R}_N$ and every $p \in \mathbb{R}^L_{++}$, the map

$$u \mapsto e^{\mathcal{U}}(p, u, R)$$

is twice continuously differentiable on Y, with

$$\frac{\partial e^{\mathcal{U}}(p, u, R)}{\partial u} > 0.$$

It follows that

$$\varepsilon^{\mathcal{U}}(p, u, R) = \frac{\partial \log e^{\mathcal{U}}(p, u, R)}{\partial \log u}, \qquad \eta^{\mathcal{U}}(p, u, R) = \frac{\partial \log \varepsilon^{\mathcal{U}}(p, u, R)}{\partial \log u}$$

are well defined, as well as the pooled curvature envelope defined below.

Definition 5.3 (Pooled curvature envelope)**.** Fix $\rho \in (-\infty, 1]$. For a given regular function $\mathcal{U}$, the pooled raw lower envelope associated with ρ is

$$a_\rho^{\mathcal{U},0}(u) := \inf_{R \in \mathcal{R}_N} \inf_{p \in \Delta} \big\{\rho\, \varepsilon^{\mathcal{U}}(p, u, R) + \eta^{\mathcal{U}}(p, u, R)\big\}$$

The pooled curvature envelope is the lower-semicontinuous envelope of this raw bound:

$$a_\rho^{\mathcal{U}}(u) := \liminf_{\substack{v \to u \\ v \in Y}} a_\rho^{\mathcal{U},0}(v), \qquad u \in Y.$$

The characterization then takes the same form as in the common-preference case, except that the curvature bound is now pooled across both prices and preference relations.

Theorem 5.4 (Characterization - unique profile version)**.** *Let $\rho \in (-\infty, 1]$, and fix a regular contour scale $\mathcal{U}$. There exists a regular $\mathcal{U}$-utilitarian social ranking that is ϕ_ρ-prioritarian if and only if the pooled curvature envelope $a_\rho^{\mathcal{U}}$ is finite-valued and continuous on Y.*

When this condition holds, the regular $\mathcal{U}$-utilitarian social rankings that are ϕ_ρ-prioritarian are exactly those represented by

$$W(x) = \sum_{i=1}^{N} \big(F_\rho^{\mathcal{U}} \circ U_{R_i}^{\mathcal{U}}\big)(x_i),$$

where $F_\rho^{\mathcal{U}}$ is constructed from $a_\rho^{\mathcal{U}}$ exactly as in Theorem 3.11.

Unlike in the identical-preference case, the transform $F^{\mathcal{U}}$ now depends on the initial choice of the interpersonal well-being function $\mathcal{U}$. Thus, ϕ-prioritarianism does not by itself solve the comparability problem: it tells us how to cardinalize and aggregate well-being once an interpersonal measure has been chosen, but it does not select that measure. In particular, $\mathcal{U}$ may itself be money-metric, with a reference price chosen for interpersonal-comparison purposes. If canonical real income also happens to admit a money-metric representation - as studied in Section 3 - the corresponding reference price need not coincide with the one embedded in $\mathcal{U}$. The two play distinct normative roles: the latter serves interpersonal comparability, whereas the former is selected by ϕ-prioritarianism so as to encode consistent inequality aversion among equal preference individuals.

The final form is very close to Piacquadio's opportunity-equivalent utilitarian social rankings of the form $x \mapsto \sum_i \phi \circ G \circ U_i(x)$. However, as discussed in section 3, for a given ϕ, the overall inequality aversion depends on the choice of the least joint concave representation G, which is *a priori* arbitrary. The ϕ-prioritarian approach circumvents this issue by pinning down $F_\rho^{\mathcal{U}}$ through the ϕ-prioritarian fairness axiom. The two approaches only coincide when $\phi_\rho(w) = w$.

This social ranking was constructed to rank allocations given the fixed preference profile $(R_1, ...R_N)$. One may, however, want a social ranking that can rank allocations given any preference profile. There is no difficulty in adapting the characterization to such a multi-profile framework, if one is willing to restrict the preference domain.

5.3 Multi-profile framework

Let $\mathcal{R}^* \subseteq \mathcal{R}$ be a subset of all continuous, convex, and monotone preferences. We now move from the unique-profile framework to a setting in which the social ranking is defined for every finite preference profile

$$R^n = (R_1, \dots, R_n) \in (\mathcal{R}^*)^n, \qquad n \geq 1.$$

A social state is therefore a pair (x, R^n), where $x = (x_1, \dots, x_n) \in X^n$. Let $\mathcal{I}^* = \{I(x,R), x \in X, R \in \mathcal{R}^*\}$ Given a contour scale $\mathcal{U} : \mathcal{I}^* \to Y$, the translation of $\mathcal{U}$-utilitarianism is immediate. A multi-profile social ranking is $\mathcal{U}$-utilitarian if there exists an increasing function $f : Y \to \mathbb{R}$ such that, for every $n \geq 1$, every profile $R^n = (R_1, \ldots, R_n) \in (\mathcal{R}^*)^n$, and every allocation $x \in X^n$,

$$W_{R^n}(x) = \sum_{i=1}^{n} f\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

represents the ranking at profile R^n. The extension of ϕ-prioritarianism to this multi-profile setting is immediate. The representation theorem then depends on the following curvature envelope, pooled over all $R \in \mathcal{R}^*$.

Definition 5.5 (Pooled curvature envelope - multi-profile version)**.** Fix $\rho \in (-\infty, 1]$. For a regular contour scale $\mathcal{U}$, the pooled raw curvature bound is

$$a_\rho^{\mathcal{U},0}(u) := \inf_{R \in \mathcal{R}^*} \inf_{p \in \Delta} \left\{ \rho\, \varepsilon^{\mathcal{U}}(p, u, R) + \eta^{\mathcal{U}}(p, u, R) \right\}.$$

The pooled curvature envelope is the lower-semicontinuous envelope of this raw bound:

$$a_\rho^{\mathcal{U}}(u) := \liminf_{\substack{v \to u \\ v \in Y}} a_\rho^{\mathcal{U},0}(v), \qquad u \in Y.$$

The characterization can finally be stated as follows.

Theorem 5.6 (Characterization - multi-profile framework)**.** *Let $\rho \in (-\infty, 1]$, and fix a regular contour scale $\mathcal{U}$. There exists a regular $\mathcal{U}$-utilitarian multi-profile social ranking that is ϕ_ρ-prioritarian if and only if the pooled curvature envelope $a_\rho^{\mathcal{U}}$ is finite-valued and continuous on Y.*

When this condition holds, the regular $\mathcal{U}$-utilitarian multi-profile social rankings that are ϕ_ρ-prioritarian are exactly those such that, for every $n \geq 1$ and every profile

$$R^n = (R_1, \ldots, R_n) \in (\mathcal{R}^*)^n,$$

the ranking at profile R^n is represented by

$$W_{R^n}(x) = \sum_{i=1}^{n} \big(F_\rho^{\mathcal{U}} \circ U_{R_i}^{\mathcal{U}}\big)(x_i),$$

where $F_\rho^{\mathcal{U}}$ is constructed from $a_\rho^{\mathcal{U}}$ exactly as in Theorem 3.11.

With $\mathcal{R}^* = \mathcal{R}$, one finds the usual result that it is impossible to construct a fair social ranking without an infinite degree of inequality aversion. The reason is that the pooled lower envelope

$$\inf_{R \in \mathcal{R}^*} \inf_{p \in \Delta} \big\{ \rho\, \varepsilon^{\mathcal{U}}(p, u, R) + \eta^{\mathcal{U}}(p, u, R) \big\}$$

is $-\infty$ for $\mathcal{R}^* = \mathcal{R}$. The impossibility can only be escaped by restricting the domain of admissible preferences. This mirrors the fixed-preference case: when a single preference relation is fixed, it is standard to impose concavifiability, since without it even the Pigou–Dalton principle can be satisfied, together with the other basic axioms, only by imposing an infinite degree of inequality aversion (Fleurbaey and Maniquet 2011).[18] The domain restriction proposed here is the multi-profile analogue of this familiar move.[19] In such a case, ϕ-prioritarianism does not say anything about how to make interpersonal comparisons, but does allow for consistent inequality aversion once an interpersonally comparable measure has been chosen.

6 Conclusion

The paper's central claim is that, under non-homothetic preferences, the welfare cardinalization and the degree of inequality aversion in social welfare aggregation cannot be chosen independently. A money-metric utility is not a neutral pre-processing step: once combined with a concave aggregator, it determines how distributive gains and losses are weighted. The contribution of ϕ-prioritarianism is to make that ethical content explicit. By encoding tolerated leakage directly in the transfer axiom, it selects a canonical welfare scale - and therefore a canonical notion of real income - that is consistent with the chosen degree of prioritarian concern. This perspective yields both a conceptual clarification and a set of tractable formulas. Conceptually, it explains why reference prices matter in growth and inflation evaluation: they are part of the distributive judgment, not merely a normalization convention. Analytically, it produces closed-form real-income corrections for standard demand systems and extends to heterogeneous-preference environments once suitable interpersonal-comparison axioms are added.

Next steps would be both theoretical and empirical. From a theoretical perspective, though the paper provides a characterization of when the canonical welfare scale admits a money-metric representation, much remains to be understood about the underlying structure of preferences that makes such a representation possible. In particular, a key open question is to identify economically meaningful conditions under which the same reference price system minimizes the relevant local curvature term across utility levels, so that the canonical representation can take the form of a money-metric utility. Put differently, one would like a deeper theory of the classes of non-homothetic preferences for which the ethical problem of welfare cardinalization still leads back to a money-metric object. This calls for relating the existence of such a reference price to primitive features of demand, such as the way expenditure elasticities and their variation with utility are jointly organized across goods.

Empirically, the formulas derived here suggest that the size of the non-homotheticity correction depends on observable demand objects, in particular on the joint behavior of budget shares and expenditure elasticities across income groups and over time. Richer expenditure data could therefore be used to estimate the implied welfare corrections more systematically, across countries, across periods, and across alternative specifications of consumer demand. This would make it possible to assess not only whether standard growth and inflation statistics are biased, but also whether those biases are quantitatively important from the perspective of explicit distributive judgments.

[18] More precisely, the 'maximin' theorem from Fleurbaey and Maniquet (2011) states that Pigou-Dalton, Pareto, and Unchanged contour independence force a maximin type social welfare ordering.

[19] However, restricting preferences to the class of concavifiable preferences is not enough to invalidate the maximin theorem, because $\inf_{R\in\mathcal{R}^*}\inf_{p\in\Delta}\left\{\rho\,\varepsilon^{\mathcal{U}}(p,u,R)+\eta^{\mathcal{U}}(p,u,R)\right\}$ is still unbounded below for that class.

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Appendices

A Extensions to Section 4

This appendix spells out the calculations behind the formulas reported in Section 4. The strategy is the same in every case. First, compute

$$\varepsilon(p,u) = \frac{\partial \log e(p,u)}{\partial \log u}, \qquad \eta(p,u) = \frac{\partial \log \varepsilon(p,u)}{\partial \log u},$$

and then minimize

$$a_\rho(p,u) := \rho\varepsilon(p,u) + \eta(p,u)$$

over prices. Second, reconstruct the canonical welfare scale $F \circ U$ from Theorem 3.11. Whenever possible, we then exhibit a canonical real-income representation U^* in the sense of Definition 3.12, that is, a utility representation satisfying

$$\phi_\rho \circ U^* \sim F \circ U.$$

A.1 PIGL

We start with the PIGL specification.[20] Consider

$$e(p,u) = \left[(1-u)\, f(p)^\kappa + u\, g(p)^\kappa\right]^{1/\kappa} = \left[A(p) + uC(p)\right]^{1/\kappa},$$

where

$$A(p) = f(p)^\kappa, \qquad B(p) = g(p)^\kappa, \qquad C(p) = B(p) - A(p) > 0.$$

It is convenient to introduce

$$c(p) := \frac{A(p)}{C(p)}.$$

Then

$$\varepsilon(p,u) = \frac{1}{\kappa}\frac{uC(p)}{A(p) + uC(p)} = \frac{1}{\kappa}\frac{u}{u + c(p)},$$

and

$$\eta(p,u) = \frac{\partial}{\partial \log u} \log\left(\frac{1}{\kappa}\frac{u}{u + c(p)}\right) = 1 - \frac{u}{u + c(p)} = \frac{c(p)}{u + c(p)}.$$

Therefore

$$a_\rho(p,u) = \rho\varepsilon(p,u) + \eta(p,u) = 1 + \frac{\rho - \kappa}{\kappa}\frac{u}{u + c(p)}.$$

For fixed u, the map $c \mapsto \frac{u}{u+c}$ is strictly decreasing, so the minimizer depends only on the sign of $\rho - \kappa$.

[20]The Gorman form is recovered as a special case of PIGL, by taking $\kappa = 1$.

Case 1: $\rho - \kappa \le 0$. Define

$$\underline{c} := \inf_p \frac{A(p)}{C(p)}.$$

Since $\rho - \kappa \le 0$, minimizing $a_\rho(p, u)$ means maximizing $\frac{u}{u+c(p)}$, hence taking $c(p)$ as small as possible. The lower envelope is therefore

$$a(u) = 1 + \frac{\rho - \kappa}{\kappa} \frac{u}{u + \underline{c}}.$$

The corresponding minimizing elasticity profile is

$$\bar{\varepsilon}(u) = \frac{1}{\kappa} \frac{u}{u + \underline{c}}, \qquad \bar{\eta}(u) = \frac{\underline{c}}{u + \underline{c}},$$

and it indeed satisfies

$$\rho \bar{\varepsilon}(u) + \bar{\eta}(u) = a(u).$$

Hence, by Theorem 3.15, the extended money-metric utility generated by $\bar{\varepsilon}$ is a canonical real-income representation:

$$U^*(x) = \exp\left(\int_{u_0}^{U(x)} \frac{1}{\kappa} \frac{t}{t + \underline{c}} \, d \log t \right) \sim \big(U(x) + \underline{c} \big)^{1/\kappa}.$$

Thus one may take

$$U^*(x) = \big(U(x) + \underline{c} \big)^{1/\kappa}.$$

When the infimum is attained at some price $\tilde{p}$, that price is a money-metric reference price for canonical real income. More generally, any sequence of prices along which $A(p_n)/C(p_n) \to \underline{c}$ generates the same extended money-metric representation.

If $\underline{c} = 0$, then $\bar{\varepsilon}(u) \equiv 1/\kappa$. Along an absolutely continuous Marshallian path $s \mapsto (p_s, x_s, I_s)$, Corollary 3.16 gives

$$\frac{d \log U^*(x_s)}{ds} = \frac{\bar{\varepsilon}(u_s)}{\varepsilon(p_s, u_s)} \operatorname{Div}(s) = \frac{\frac{1}{\kappa}}{\frac{1}{\kappa} \frac{u_s C(p_s)}{A(p_s) + u_s C(p_s)}} \operatorname{Div}(s).$$

Since $I_s^\kappa = A(p_s) + u_s C(p_s)$, this becomes

$$\frac{d \log U^*(x_s)}{ds} = \frac{I_s^\kappa}{I_s^\kappa - A(p_s)} \operatorname{Div}(s),$$

hence

$$U^*(x_t) = U^*(x_0) \exp\left(\int_0^t \frac{I_s^\kappa}{I_s^\kappa - A(p_s)} \operatorname{Div}(s) \, ds \right).$$

Case 2: $\rho - \kappa > 0$**.** Now minimizing $a_\rho(p,u)$ means minimizing $\frac{u}{u+c(p)}$, hence maximizing $c(p)$. Let

$$\bar{c} := \sup_p \frac{A(p)}{C(p)}.$$

If $\bar{c} < +\infty$, the same calculation as above gives

$$\bar{\varepsilon}(u) = \frac{1}{\kappa}\frac{u}{u+\bar{c}},$$

so the corresponding money-metric canonical real-income representation is

$$U^*(x) \sim \big(U(x) + \bar{c}\big)^{1/\kappa}.$$

Thus one may take

$$U^*(x) = \big(U(x) + \bar{c}\big)^{1/\kappa}.$$

The associated reference price system is any price (or price sequence) that attains the supremum $\bar{c}$.

If instead $\bar{c} = +\infty$, then $c(p) \to +\infty$ forces $\varepsilon(p,u) \to 0$, so

$$a(u) = \inf_p a_\rho(p,u) = 1.$$

The canonical welfare scale $F \circ U$ is then affine in U. Since $\rho \neq 0$, a canonical real-income representation exists and may be chosen as

$$U^*(x) \sim U(x)^{1/\rho},$$

because

$$\phi_\rho\big(U(x)^{1/\rho}\big) = \frac{U(x)}{\rho}$$

is affine in $U(x)$. In this case there is no money-metric representation, because every minimizing sequence drives $\varepsilon(p,u)$ to 0, whereas Theorem 3.15 requires a strictly positive limiting elasticity profile.

Example (Boppart's PIGL subclass). Consider

$$V(p,I) = \frac{1}{\kappa}\left[\left(\frac{I}{B(p)}\right)^{\kappa} - 1\right] - \frac{\nu}{\gamma}\left[\left(\frac{D(p)}{B(p)}\right)^{\gamma} - 1\right].$$

Solving for I gives

$$I^\kappa = B(p)^\kappa\left[\kappa V + 1 - \frac{\kappa\nu}{\gamma} + \frac{\kappa\nu}{\gamma}\left(\frac{D(p)}{B(p)}\right)^{\gamma}\right].$$

Hence this is a PIGL form with

$$A(p) = B(p)^\kappa\left[1 - \frac{\kappa\nu}{\gamma} + \frac{\kappa\nu}{\gamma}\left(\frac{D(p)}{B(p)}\right)^{\gamma}\right], \qquad C(p) = \kappa B(p)^\kappa.$$

Therefore

$$\frac{A(p)}{C(p)} = \frac{1}{\kappa} - \frac{\nu}{\gamma} + \frac{\nu}{\gamma}\left(\frac{D(p)}{B(p)}\right)^{\gamma}.$$

So

$$\inf_p \frac{A(p)}{C(p)} = \frac{1}{\kappa} - \frac{\nu}{\gamma}, \qquad \sup_p \frac{A(p)}{C(p)} = +\infty.$$

The general PIGL formulas immediately imply:

- if $\rho - \kappa \le 0$, a canonical real-income representation is

$$V^*(p,I) = \left[\frac{1}{\kappa}\left(\frac{I}{B(p)}\right)^{\kappa} - \frac{\nu}{\gamma}\left(\frac{D(p)}{B(p)}\right)^{\gamma}\right]^{1/\kappa};$$

- if $\rho - \kappa > 0$, the canonical welfare scale is affine in V, so one may take

$$V^*(p,I) \sim V(p,I)^{1/\rho}.$$

A.2 PIGLOG

The PIGLOG form is the limit of PIGL when $\kappa \to 0$. Write

$$\log e(p,u) = (1-u)\log f(p) + u \log g(p) = A(p) + uC(p),$$

where

$$A(p) = \log f(p), \qquad B(p) = \log g(p), \qquad C(p) = B(p) - A(p) > 0.$$

Then

$$\varepsilon(p,u) = \frac{\partial \log e(p,u)}{\partial \log u} = uC(p), \qquad \eta(p,u) = \frac{\partial \log(uC(p))}{\partial \log u} = 1.$$

Hence

$$a_\rho(p,u) = 1 + \rho u\, C(p).$$

Case 1: $\rho < 0$. Let

$$\bar{c} := \sup_p C(p).$$

Because $\rho < 0$, minimizing $a_\rho(p,u)$ means maximizing $C(p)$. If $\bar{c} = +\infty$, then $a(u) = -\infty$ for every $u > 0$, so the finiteness condition in Theorem 3.11 fails and no ϕ_ρ-prioritarian ordering exists.

If $\bar{c} < +\infty$, then

$$\bar{\varepsilon}(u) = \bar{c}\, u, \qquad \bar{\eta}(u) = 1,$$

and

$$\rho\bar{\varepsilon}(u) + \bar{\eta}(u) = 1 + \rho\bar{c}\, u = a(u).$$

By Theorem 3.15, a canonical real-income representation is therefore given by

$$U^*(x) = \exp\left(\int_{u_0}^{U(x)} \bar{c}\, t\, d\log t\right) \sim \exp\bigl(\bar{c}\, U(x)\bigr).$$

So one may take
$$U^*(x) = \exp\bigl(\bar{c}\, U(x)\bigr).$$
The relevant reference price is any $\tilde{p}$ such that $C(\tilde{p}) = \bar{c}$, or any sequence approaching that supremum.

Case 2: $\rho = 0$**.** Now
$$a(u) = \inf_p \eta(p,u) = 1.$$
Hence the canonical welfare scale $F \circ U$ is affine in U. A canonical real-income representation therefore exists and may be chosen as
$$U^*(x) = \exp(U(x)),$$
since
$$\log U^*(x) = U(x)$$
is affine in the canonical welfare scale.

In this case every money-metric utility is a canonical real-income representation: for any fixed reference price $\tilde{p}$,
$$M_{\tilde{p}}(x) = \exp\bigl(C(\tilde{p})\, U(x)\bigr),$$
hence
$$\log M_{\tilde{p}}(x) = C(\tilde{p})\, U(x),$$
which is a positive affine transform of $U(x)$.

Case 3: $\rho > 0$**.** Let
$$\underline{c} := \inf_p C(p).$$
Because $\rho > 0$, minimizing $a_\rho(p,u)$ means minimizing $C(p)$. If $\underline{c} > 0$, then
$$\bar{\varepsilon}(u) = \underline{c}\, u, \qquad \bar{\eta}(u) = 1,$$
so by Theorem 3.15 a canonical real-income representation is
$$U^*(x) \sim \exp\bigl(\underline{c}\, U(x)\bigr).$$
Thus one may take
$$U^*(x) = \exp\bigl(\underline{c}\, U(x)\bigr).$$
The money-metric reference price is any price (or price sequence) that attains $\underline{c}$.

If $\underline{c} = 0$, then the minimizing sequence forces $\varepsilon(p,u) \to 0$, so
$$a(u) = 1.$$
The canonical welfare scale is affine in U. Since $\rho > 0$, one may therefore take
$$U^*(x) \sim U(x)^{1/\rho},$$
because $\phi_\rho(U(x)^{1/\rho}) = U(x)/\rho$ is affine in $U(x)$. Again, there is no money-metric representation because the minimizing elasticity profile collapses to 0.

Example (AIDS). For the Almost Ideal Demand System,

$$\log f(p) = \alpha_0 + \sum_i \alpha_i \log p_i + \frac{1}{2} \sum_i \sum_j \gamma_{ij} \log p_i \log p_j,$$

and

$$\log g(p) = \log f(p) + \sum_i \beta_i \log p_i.$$

Therefore

$$C(p) = \sum_i \beta_i \log p_i.$$

On the price domain relevant for the AIDS specification, $C(p)$ can be made arbitrarily close to 0 and arbitrarily large, so

$$\inf_p C(p) = 0, \qquad \sup_p C(p) = +\infty.$$

The previous case distinction then gives:

- if $\rho > 0$, a canonical real-income representation is $U^*(x) \sim U(x)^{1/\rho}$;
- if $\rho = 0$, one may take $U^*(x) = \exp(U(x))$;
- if $\rho < 0$, the finiteness condition fails and no ϕ_ρ-prioritarian ordering exists.

A.3 nh-CES

We now allow both $\sigma < 1$ and $\sigma > 1$. The knife-edge case $\sigma = 1$ is omitted, since the closed-form expenditure function used below is singular there and must be treated as a separate limit. For $\sigma \neq 1$, the expenditure function is

$$e(p, u) = \left(\sum_i \Omega_i p_i^{1-\sigma} u^{(1-\sigma)\epsilon_i} \right)^{1/(1-\sigma)}.$$

Let

$$w_i(p, u) = \frac{\Omega_i p_i^{1-\sigma} u^{(1-\sigma)\epsilon_i}}{\sum_j \Omega_j p_j^{1-\sigma} u^{(1-\sigma)\epsilon_j}}$$

denote Hicksian budget shares, and write

$$m(p, u) := \sum_i w_i(p, u) \epsilon_i.$$

Differentiating $\log e$ with respect to $\log u$ gives

$$\varepsilon(p, u) = m(p, u) = \mathbb{E}_w(\epsilon).$$

Next,

$$\frac{\partial w_i(p, u)}{\partial \log u} = (1 - \sigma) w_i(p, u) \big(\epsilon_i - m(p, u) \big),$$

so

$$\frac{\partial m(p,u)}{\partial \log u} = (1-\sigma)\sum_i w_i(p,u)\big(\epsilon_i - m(p,u)\big)\epsilon_i = (1-\sigma)\mathbb{V}_w(\epsilon).$$

Therefore

$$\eta(p,u) = \frac{1}{m(p,u)}\frac{\partial m(p,u)}{\partial \log u} = (1-\sigma)\frac{\mathbb{V}_w(\epsilon)}{\mathbb{E}_w(\epsilon)}.$$

The quantity to minimize is thus

$$a_\rho(p,u) = \rho\,\mathbb{E}_w(\epsilon) + (1-\sigma)\frac{\mathbb{V}_w(\epsilon)}{\mathbb{E}_w(\epsilon)}.$$

A useful observation is that, for each fixed u, varying prices is equivalent to varying the share vector w over the simplex. Indeed, given any interior w, define

$$p_i(w,u) \propto \left(\frac{w_i}{\Omega_i u^{(1-\sigma)\epsilon_i}}\right)^{1/(1-\sigma)};$$

after normalization to Δ, this price vector generates exactly the Hicksian shares w. Therefore the infimum over prices can be computed as an infimum over share vectors.

Case 1: $\sigma < 1$. Now $1-\sigma > 0$, so the variance term is nonnegative.

If $\rho < 0$, then

$$a_\rho(w) = \rho\,\mathbb{E}_w(\epsilon) + (1-\sigma)\frac{\mathbb{V}_w(\epsilon)}{\mathbb{E}_w(\epsilon)} \geq \rho\,\mathbb{E}_w(\epsilon) \geq \rho\,\epsilon_L,$$

with equality only when all weight is on the most luxurious good L. Hence

$$a(u) = \rho\,\epsilon_L \qquad (\rho < 0).$$

If $\rho = 0$, then

$$a_0(w) = (1-\sigma)\frac{\mathbb{V}_w(\epsilon)}{\mathbb{E}_w(\epsilon)} \geq 0,$$

and the minimum 0 is attained by any degenerate share vector. Thus

$$a(u) = 0 \qquad (\rho = 0).$$

If $\rho > 0$, then

$$a_\rho(w) \geq \rho\,\mathbb{E}_w(\epsilon) \geq \rho\,\epsilon_1,$$

with equality only when all weight is on the least luxurious good 1. Hence

$$a(u) = \rho\,\epsilon_1 \qquad (\rho > 0).$$

Since the envelope is constant, the canonical welfare scale is explicitly determined. A canonical real-income representation may be chosen as

$$U^*(x) \sim \begin{cases} U(x)^{\epsilon_L}, & \rho < 0, \\ U(x), & \rho = 0, \\ U(x)^{\epsilon_1}, & \rho > 0, \end{cases}$$

because

$$\phi_\rho\big(U(x)^{\epsilon_L}\big) = \frac{U(x)^{\rho\epsilon_L}}{\rho}, \qquad \phi_\rho\big(U(x)^{\epsilon_1}\big) = \frac{U(x)^{\rho\epsilon_1}}{\rho},$$

and when $\rho = 0$, $\phi_\rho(U(x)) = \log U(x)$ up to the canonical normalization used in the main text. This recovers the formula stated there for $\rho \le 0$, and shows that the $\rho > 0$ case simply selects the opposite boundary good.

In these boundary cases the money-metric reference price system is also immediate:

- if $\rho < 0$, put all weight on good L: $p_L = 1$, $p_i = 0$ for $i \neq L$;
- if $\rho > 0$, put all weight on good 1: $p_1 = 1$, $p_i = 0$ for $i \neq 1$;
- if $\rho = 0$, any degenerate boundary price concentrating on a single good works.

Case 2: $\sigma > 1$. Set $\lambda := \sigma - 1 > 0$. Then

$$a_\rho(w) = \rho\,\mathbb{E}_w(\epsilon) - \lambda\frac{\mathbb{V}_w(\epsilon)}{\mathbb{E}_w(\epsilon)}.$$

For a fixed mean $m \in [\epsilon_1, \epsilon_L]$, the second term is decreasing in the variance, so the minimizing share vector is the one with maximal variance conditional on that mean. This is obtained by putting weight only on the two extreme elasticities ϵ_1 and ϵ_L. For such a two-point distribution, the variance is

$$\mathbb{V}_w(\epsilon) = (\epsilon_L - m)(m - \epsilon_1).$$

Hence the problem reduces to

$$a(u) = \min_{m \in [\epsilon_1, \epsilon_L]} g_\rho(m),$$

where

$$g_\rho(m) = \rho m - \lambda\frac{(\epsilon_L - m)(m - \epsilon_1)}{m} = (\rho + \lambda)m - \lambda(\epsilon_1 + \epsilon_L) + \lambda\frac{\epsilon_1\epsilon_L}{m}.$$

Since

$$g_\rho''(m) = 2\lambda\frac{\epsilon_1\epsilon_L}{m^3} > 0,$$

g_ρ is strictly convex. Its unconstrained minimizer is

$$m_\rho^\star = \sqrt{\frac{\lambda\epsilon_1\epsilon_L}{\rho + \lambda}}$$

whenever $\rho + \lambda > 0$; if $\rho + \lambda \le 0$, g_ρ is strictly decreasing and the minimum is attained at $m = \epsilon_L$.

It is convenient to define the thresholds

$$\rho_L := \lambda\left(\frac{\epsilon_1}{\epsilon_L} - 1\right) < 0, \qquad \rho_1 := \lambda\left(\frac{\epsilon_L}{\epsilon_1} - 1\right) > 0.$$

Then the minimizer is

$$m^\star(\rho) = \begin{cases} \epsilon_L, & \rho \le \rho_L, \\ \sqrt{\dfrac{\lambda\epsilon_1\epsilon_L}{\rho+\lambda}}, & \rho_L < \rho < \rho_1, \\ \epsilon_1, & \rho \ge \rho_1. \end{cases}$$

Accordingly,

$$a(u) = \begin{cases} \rho\,\epsilon_L, & \rho \le \rho_L, \\ 2\sqrt{\lambda(\rho+\lambda)\epsilon_1\epsilon_L} - \lambda(\epsilon_1+\epsilon_L), & \rho_L < \rho < \rho_1, \\ \rho\,\epsilon_1, & \rho \ge \rho_1. \end{cases}$$

When $\rho \ne 0$, the envelope is constant, so the canonical welfare scale is affine to U^a. A canonical real-income representation may therefore be chosen as

$$U^*(x) \sim U(x)^{a/\rho}.$$

Thus

$$U^*(x) \sim \begin{cases} U(x)^{\epsilon_L}, & \rho \le \rho_L, \\ U(x)^{\frac{2\sqrt{\lambda(\rho+\lambda)\epsilon_1\epsilon_L} - \lambda(\epsilon_1+\epsilon_L)}{\rho}}, & \rho_L < \rho < \rho_1,\ \rho \ne 0, \\ U(x)^{\epsilon_1}, & \rho \ge \rho_1. \end{cases}$$

At $\rho = 0$, which always lies in the interior region (ρ_L, ρ_1), we obtain

$$a_0 = 2\lambda\sqrt{\epsilon_1\epsilon_L} - \lambda(\epsilon_1+\epsilon_L) = -\lambda\big(\sqrt{\epsilon_L} - \sqrt{\epsilon_1}\big)^2 < 0.$$

The canonical welfare scale $F \circ U$ is then affine to U^{a_0}/a_0. Hence a canonical real-income representation may be chosen as

$$U^*(x) = \exp\left(\frac{U(x)^{a_0}}{a_0}\right),$$

since

$$\log U^*(x) = \frac{U(x)^{a_0}}{a_0}$$

is affine in the canonical welfare scale.

The money-metric interpretation is also clear. In the two boundary regions ($\rho \le \rho_L$ and $\rho \ge \rho_1$), the minimizing shares are degenerate, so the same boundary reference prices as in the $\sigma < 1$ case apply. In the interior region $\rho_L < \rho < \rho_1$, however, the minimizing share vector is a nondegenerate two-good mix between goods 1 and L. No single reference price system can generate that mix for all utility levels, because for any fixed two-good support one has

$$\frac{w_1(p,u)}{w_L(p,u)} = \frac{\Omega_1 p_1^{1-\sigma}}{\Omega_L p_L^{1-\sigma}}\, u^{(1-\sigma)(\epsilon_1-\epsilon_L)},$$

which necessarily varies with u unless the limit is degenerate. Hence the interior $\sigma > 1$ cases generally do *not* admit a money-metric representation.

A.4 CEX application with no equivalization

The graphs below show the results where no equivalization is done on expenditure. The non-homotheticity correction changes slightly quantitatively, but the Engel curves are almost identical.

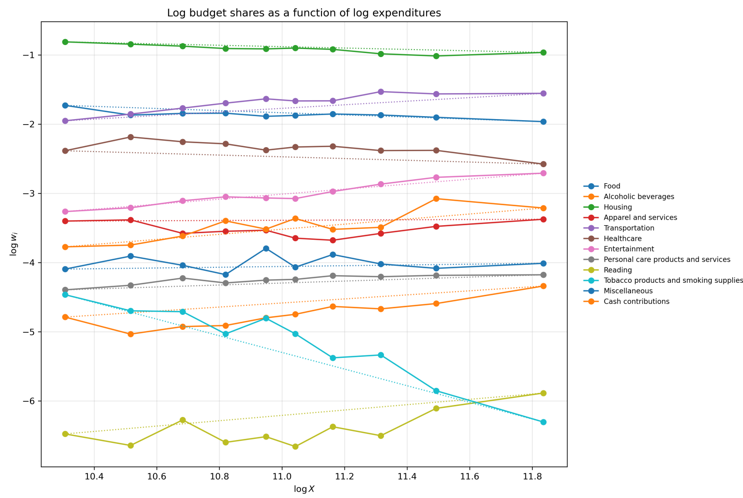

Figure 10: Engel curves with no equivalization

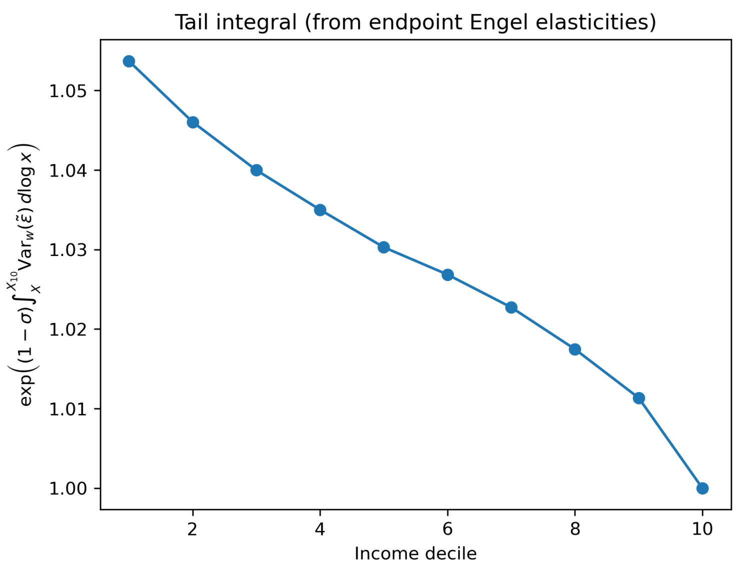

Figure 11: Non-homotheticity correction with no equivalization

B Proofs

B.1 Proof of the prioritarian social-ordering lemma 3.5

Proof. This result is well known. By continuity and Separability, the Debreu-Gorman additive-representation theorem yields continuous one-person indices $\psi_i : X \to \mathbb{R}$ such that

$$(x_1, \ldots, x_N) \succsim (y_1, \ldots, y_N) \iff \sum_{i=1}^{N} \psi_i(x_i) \geq \sum_{i=1}^{N} \psi_i(y_i).$$

Pareto implies that each ψ_i is strictly increasing along the common preference ordering. Hence, on the common utility range $Y := U(X)$, there exist continuous increasing maps $f_i : Y \to \mathbb{R}$ with

$$\psi_i = f_i \circ U \qquad (i = 1, \ldots, N).$$

One can then invoke the classical characterization of continuous separable social orderings on identical domains that satisfy Pigou-Dalton: the one-person terms must coincide up to additive constants, and their common part must be concave; see, e.g., Gorman (1959). Therefore there exist a continuous increasing concave function $\psi : X \to \mathbb{R}$ and constants c_i such that

$$\psi_i = \psi + c_i \qquad (i = 1, \ldots, N).$$

Since the additive constants cancel from social comparisons, the ranking is represented by

$$W(x) = \sum_{i=1}^{N} \psi(x_i).$$

Now let U^{LC} be any least concave representation of the common preferences; existence is standard, see Debreu (1976) or Kannai (1977). Because ψ is itself a concave representation of the same preferences, the defining minimality property of U^{LC} implies that there exists an increasing concave map ϕ such that

$$\psi = \phi \circ U^{LC}.$$

Hence

$$W(x) = \sum_{i=1}^{N} \phi(U^{LC}(x_i)).$$

Conversely, any ranking of the displayed form is continuous and separable. Pareto holds because both U^{LC} and ϕ are increasing. Pigou-Dalton holds because the one-person term $\phi \circ U^{LC}$ is concave on X. □

B.2 Proof of the Pigou-Dalton equivalence lemma 3.7

It is convenient to prove the following equivalent formulation.

Lemma B.1. *Let the social ordering be represented by*

$$W(x) = \sum_{i=1}^{N} \psi(x_i),$$

where $\psi : X \to \mathbb{R}$ is continuous and strictly increasing. Then the following are equivalent:

(i) the ordering satisfies bundle Pigou-Dalton;

(ii) ψ is concave on X;

(iii) for every strictly positive price vector p, the indirect one-person welfare function

$$\chi_p(I) := \psi(x^*(p, I))$$

is concave in income, and therefore the ordering satisfies income Pigou-Dalton.

Proof. **(ii) ⇒ (i).** Let a bundle Pigou-Dalton transfer replace (x_i, x_j) by

$$x_i' = (1-\delta)x_i + \delta x_j, \qquad x_j' = \delta x_i + (1-\delta)x_j, \qquad 0 < \delta < \tfrac{1}{2}.$$

Concavity of ψ gives

$$\psi(x_i') \geq (1-\delta)\psi(x_i) + \delta\psi(x_j), \qquad \psi(x_j') \geq \delta\psi(x_i) + (1-\delta)\psi(x_j).$$

Adding the two inequalities yields

$$\psi(x_i') + \psi(x_j') \geq \psi(x_i) + \psi(x_j).$$

All other terms are unchanged, so social welfare weakly rises.

(i) ⇒ (ii). Fix $x, y \in X$. For every $\delta \in (0, 1/2)$, applying bundle Pigou-Dalton to the pair (x, y) gives

$$\psi((1-\delta)x + \delta y) + \psi(\delta x + (1-\delta)y) \geq \psi(x) + \psi(y).$$

Letting $\delta \uparrow 1/2$ and using continuity of ψ, we obtain

$$2\psi\left(\frac{x+y}{2}\right) \geq \psi(x) + \psi(y).$$

Thus ψ is midpoint-concave. Since ψ is continuous, midpoint concavity implies full concavity on the convex set X.

(ii) ⇒ (iii). Fix $p \gg 0$. Because ψ and U represent the same preference ordering and ψ is increasing, the Marshallian demand maximizes ψ on the budget set:

$$\chi_p(I) = \max\{\psi(x) : px \leq I\}.$$

Let $I, J > 0$ and $\lambda \in [0, 1]$. Choose $x \in \arg\max\{\psi(z) : pz \leq I\}$ and $y \in \arg\max\{\psi(z) : pz \leq J\}$. Then $\lambda x + (1-\lambda)y$ is feasible at income $\lambda I + (1-\lambda)J$, so concavity of ψ yields

$$\chi_p(\lambda I + (1-\lambda)J) \geq \psi(\lambda x + (1-\lambda)y) \geq \lambda\psi(x) + (1-\lambda)\psi(y) = \lambda\chi_p(I) + (1-\lambda)\chi_p(J).$$

Hence χ_p is concave. For an additive social welfare function on $\mathbb{R}_+^N$, concavity of each one-person indirect term is exactly the income Pigou-Dalton property.

(iii) $\Rightarrow$ (ii). This is a standard result from consumer duality (Diewert 1978): under the maintained convexity, monotonicity, continuity, and existence assumptions on the consumer problem, concavity of the indirect representation in income for every strictly positive price vector is equivalent to concavity of the corresponding direct representation.

The three properties are therefore equivalent. In particular, for a prioritarian ranking the bundle and income versions of Pigou-Dalton coincide. □

B.3 Proof of Lemma 3.9

Proof. Let the social ranking be represented by

$$W(x) = \sum_{i=1}^{N} F(U(x_i)),$$

where F is increasing and C^2. For each price vector p, define

$$\chi_p(I) := F(V(p, I)), \qquad f_p := \chi_p \circ \phi^{-1}.$$

By definition, $P(\phi)$ says that whenever two incomes are transformed into a pair that is closer together and whose ϕ-sum is weakly larger, the sum of one-person welfare levels weakly rises. Since χ_p is increasing, this is equivalent in one dimension to the concavity of

$$f_p := \chi_p \circ \phi^{-1}$$

on $\phi(\mathbb{R}_{++})$: concavity gives the required improvement for mean-preserving contractions in ϕ-space, and monotonicity gives it for weak increases in the ϕ-sum.

Since χ_p and ϕ are C^2 and $\phi' > 0$, concavity of f_p is equivalent to

$$(\chi_p \circ \phi^{-1})'' \leq 0.$$

Writing $z = \phi(I)$, a direct differentiation gives

$$(\chi_p \circ \phi^{-1})''(z) = \frac{\chi_p''(I)\phi'(I) - \chi_p'(I)\phi''(I)}{(\phi'(I))^3},$$

so the inequality above is equivalent to

$$-\frac{I\chi_p''(I)}{\chi_p'(I)} \geq -\frac{I\phi''(I)}{\phi'(I)}. \tag{1}$$

Define

$$c(I) := \inf_{p \in \mathbb{R}_{++}^L} \left\{ -\frac{I\chi_p''(I)}{\chi_p'(I)} \right\}.$$

Then $P(\phi)$ holds if and only if

$$-\frac{I\phi''(I)}{\phi'(I)} \leq c(I) \qquad \forall I > 0. \tag{2}$$

We now show that c is constant. Since indirect utility is homogeneous of degree zero in (p, I), one has

$$\chi_{\lambda p}(\lambda I) = \chi_p(I) \qquad \forall \lambda > 0.$$

Differentiating with respect to I yields

$$\chi'_{\lambda p}(\lambda I) = \lambda^{-1}\chi'_p(I), \qquad \chi''_{\lambda p}(\lambda I) = \lambda^{-2}\chi''_p(I).$$

Hence

$$-\frac{\lambda I\, \chi''_{\lambda p}(\lambda I)}{\chi'_{\lambda p}(\lambda I)} = -\frac{I\, \chi''_p(I)}{\chi'_p(I)}.$$

Taking the infimum over p gives

$$c(\lambda I) = c(I) \qquad \forall \lambda > 0,\ \forall I > 0.$$

Therefore c is constant on $\mathbb{R}_{++}$, and (2) becomes

$$-\frac{I\phi''(I)}{\phi'(I)} \leq c \qquad \forall I > 0. \tag{3}$$

Set

$$c_\phi := \sup_{I>0} \left\{ -\frac{I\phi''(I)}{\phi'(I)} \right\}.$$

Since $P(\phi)$ holds for the present ranking, (3) implies $c_\phi \leq c$.

Now let $\rho := 1 - c_\phi \leq 1$, and let ϕ_ρ be any member of the isoelastic family:

$$\phi_\rho(w) = \begin{cases} \dfrac{w^\rho}{\rho}, & \rho \neq 0, \\ \log w, & \rho = 0, \end{cases}$$

up to a positive affine transformation. By construction,

$$-\frac{I\phi''_\rho(I)}{\phi'_\rho(I)} = 1 - \rho = c_\phi \qquad \forall I > 0. \tag{4}$$

Combining $c_\phi \leq c$ with (4), we obtain

$$-\frac{I\phi''_\rho(I)}{\phi'_\rho(I)} \leq c = c(I) \qquad \forall I > 0.$$

By the equivalence established above, this implies that $P(\phi_\rho)$ holds. Hence the given social ranking is ϕ_ρ-admissible.

It remains to prove the minimality part. Let $\succsim'$ be any ϕ_ρ-admissible prioritarian social ranking. Repeating the previous argument for $\succsim'$, there exists a constant c' such that $\succsim'$ is ϕ_ρ-admissible if and only if

$$-\frac{I\phi''_\rho(I)}{\phi'_\rho(I)} \le c' \qquad \forall I > 0.$$

Using (4), this is equivalent to

$$c_\phi \le c'.$$

But then, since

$$-\frac{I\phi''(I)}{\phi'(I)} \le c_\phi \le c' \qquad \forall I > 0,$$

the same equivalence shows that $\succsim'$ is also ϕ-admissible.

Now the original ranking is ϕ-prioritarian, so by definition it is among the least inequality-averse ϕ-admissible social rankings: every equal-price equalizing transfer endorsed by it is also endorsed by any other ϕ-admissible social ranking. Since every ϕ_ρ-admissible ranking is also ϕ-admissible, it follows in particular that every equal-price equalizing transfer endorsed by the original ranking is also endorsed by any ϕ_ρ-admissible ranking. Therefore the original ranking is ϕ_ρ-prioritarian.

This proves the result. □

B.4 Proof of the existence and canonical-representation theorem 3.11

B.4.1 Auxiliary comparison lemmas

Lemma B.2 (Transfer inclusion under concave reparametrization)**.** *Let $J \subset \mathbb{R}$ be an interval, let $f, g : J \to \mathbb{R}$ be increasing functions, and suppose that*

$$g = h \circ f$$

for some increasing concave function h on $f(J)$. Let $a < b$ and let a', b' satisfy

$$a \le a' \le b' \le b.$$

If

$$f(a') + f(b') \ge f(a) + f(b),$$

then

$$g(a') + g(b') \ge g(a) + g(b).$$

Proof. Let $A = f(a)$, $B = f(b)$, $A' = f(a')$, and $B' = f(b')$. Then $A \le A' \le B' \le B$ and $A' + B' \ge A + B$. Define

$$A'' := A + B - B'.$$

Then $A \le A'' \le A' \le B' \le B$. The pair (A'', B') is a mean-preserving contraction of (A, B), so concavity of h gives

$$h(A'') + h(B') \ge h(A) + h(B).$$

Since h is increasing and $A' \geq A''$,

$$h(A') + h(B') \geq h(A'') + h(B') \geq h(A) + h(B).$$

Thus $g(a') + g(b') \geq g(a) + g(b)$. $\square$

Lemma B.3 (Local separation of transfer sets)**.** *Let $J \subset \mathbb{R}$ be an open interval and let $f, g : J \to \mathbb{R}$ be increasing, concave, and C^2. Suppose that at some $z_0 \in J$,*

$$-\frac{f''(z_0)}{f'(z_0)} > -\frac{g''(z_0)}{g'(z_0)}.$$

Then there exist $\kappa > 0$ and $h_0 > 0$ such that for every $h \in (0, h_0)$, writing $s = \kappa h^2$, one has $0 < s < h$ and

$$2f(z_0 - s) - f(z_0 - h) - f(z_0 + h) > 0,$$

while

$$2g(z_0 - s) - g(z_0 - h) - g(z_0 + h) < 0.$$

Equivalently, for all sufficiently small h, the equalizing transfer from $(z_0 - h, z_0 + h)$ to $(z_0 - s, z_0 - s)$ is endorsed by f but not by g.

Proof. Choose κ such that

$$-\frac{g''(z_0)}{2g'(z_0)} < \kappa < -\frac{f''(z_0)}{2f'(z_0)}.$$

For $s = \kappa h^2$, a second-order Taylor expansion at z_0 gives

$$2f(z_0 - s) - f(z_0 - h) - f(z_0 + h) = \big(-2\kappa f'(z_0) - f''(z_0)\big)h^2 + o(h^2),$$

and similarly

$$2g(z_0 - s) - g(z_0 - h) - g(z_0 + h) = \big(-2\kappa g'(z_0) - g''(z_0)\big)h^2 + o(h^2).$$

By the choice of κ, the first coefficient is strictly positive and the second is strictly negative. Hence the two displayed expressions have the announced signs for all sufficiently small $h > 0$. Since $s = \kappa h^2 < h$ for all sufficiently small h, the move from $(z_0 - h, z_0 + h)$ to $(z_0 - s, z_0 - s)$ is indeed an equalizing transfer. $\square$

Lemma B.4 (Admissibility and the curvature envelope)**.** *Let $Y := U(X) \subset \mathbb{R}_{++}$. An increasing C^2 transform F is ϕ_ρ-admissible if and only if*

$$1 + u\frac{F''(u)}{F'(u)} \leq a_\rho(u) \qquad \forall u \in Y,$$

where a_ρ is the curvature envelope of Definition 3.10.

Proof. For each p, set

$$\chi_p(I) := F(V(p, I)).$$

From the proof of Lemma 3.9, property $\mathcal{P}(\phi_\rho)$ is equivalent to concavity of $\chi_p \circ \phi_\rho^{-1}$ for every p. Because

$$-\frac{I\phi_\rho''(I)}{\phi_\rho'(I)} = 1-\rho,$$

that concavity condition is equivalent to

$$-\frac{I\chi_p''(I)}{\chi_p'(I)} \geq 1-\rho. \tag{1}$$

Let $u := V(p,I)$, so $I = e(p,u)$. Since

$$\frac{dI}{du} = \frac{I\varepsilon(p,u)}{u}, \qquad \frac{du}{dI} = \frac{u}{I\varepsilon(p,u)},$$

one gets

$$\chi_p'(I) = F'(u)\frac{u}{I\varepsilon(p,u)}.$$

Taking logarithmic derivatives with respect to I gives

$$\frac{I\chi_p''(I)}{\chi_p'(I)} = \frac{1+uF''(u)/F'(u)-\eta(p,u)}{\varepsilon(p,u)} - 1. \tag{2}$$

Substituting (2) into (1) yields

$$1+u\frac{F''(u)}{F'(u)} \leq \rho\varepsilon(p,u)+\eta(p,u) \qquad \forall p,\ \forall u \in Y. \tag{3}$$

Equivalently, with

$$a_\rho^0(u) := \inf_{p\in\Delta}\{\rho\varepsilon(p,u)+\eta(p,u)\},$$

condition (3) is

$$A_F(u) := 1+u\frac{F''(u)}{F'(u)} \leq a_\rho^0(u) \qquad \forall u \in Y. \tag{4}$$

Since A_F is continuous and $a_\rho(u) = \liminf_{v\to u,\, v\in Y} a_\rho^0(v)$, (4) implies

$$A_F(u) = \liminf_{v\to u} A_F(v) \leq \liminf_{v\to u} a_\rho^0(v) = a_\rho(u).$$

Conversely, $a_\rho \leq a_\rho^0$ pointwise, so $A_F \leq a_\rho$ implies $A_F \leq a_\rho^0$. Thus (4) is equivalent to $A_F \leq a_\rho$, proving the claim. $\square$

Proof of Theorem 3.11. Assume first that the curvature envelope a_ρ is finite-valued and continuous on Y. Fix $u_0 \in Y$ and define

$$A_\rho(u) := \int_{u_0}^u a_\rho(s)\, d\log s, \qquad F^\star(u) := \int_{u_0}^u e^{A_\rho(t)}\, d\log t.$$

Then $F^\star$ is C^2, strictly increasing, and

$$F^{\star\prime}(u) = \frac{e^{A_\rho(u)}}{u}.$$

Hence

$$\frac{d \log F^{\star\prime}(u)}{d \log u} = a_\rho(u) - 1,$$

that is,

$$1 + u\frac{F^{\star\prime\prime}(u)}{F^{\star\prime}(u)} = a_\rho(u). \tag{5}$$

By Lemma B.4, $F^\star$ is ϕ_ρ-admissible.

We next show that the induced ranking

$$W^\star(x) := \sum_i F^\star(U(x_i))$$

is least inequality averse among the ϕ_ρ-admissible prioritarian rankings. Let

$$\widetilde{W}(x) = \sum_i \widetilde{F}(U(x_i))$$

be any other ϕ_ρ-admissible prioritarian ranking, with $\widetilde{F}$ increasing and C^2. By Lemma B.4,

$$1 + u\frac{\widetilde{F}''(u)}{\widetilde{F}'(u)} \le a_\rho(u) = 1 + u\frac{F^{\star\prime\prime}(u)}{F^{\star\prime}(u)}. \tag{6}$$

Fix a common price vector p and define

$$\chi_p^\star(I) := F^\star(V(p,I)), \qquad \widetilde{\chi}_p(I) := \widetilde{F}(V(p,I)),$$

and, in ϕ_ρ-space,

$$f_p^\star := \chi_p^\star \circ \phi_\rho^{-1}, \qquad \widetilde{f}_p := \widetilde{\chi}_p \circ \phi_\rho^{-1}.$$

Using formula (2), inequality (6) implies

$$-\frac{I\widetilde{\chi}_p''(I)}{\widetilde{\chi}_p'(I)} \ge -\frac{I\chi_p^{\star\prime\prime}(I)}{\chi_p^{\star\prime}(I)} \qquad \forall I > 0. \tag{7}$$

Since

$$-\frac{(\chi \circ \phi_\rho^{-1})''(\phi_\rho(I))}{(\chi \circ \phi_\rho^{-1})'(\phi_\rho(I))} = \frac{-I\chi''(I)/\chi'(I) + I\phi_\rho''(I)/\phi_\rho'(I)}{I\phi_\rho'(I)},$$

comparison (7) implies

$$-\frac{\widetilde{f}_p''(z)}{\widetilde{f}_p'(z)} \ge -\frac{f_p^{\star\prime\prime}(z)}{f_p^{\star\prime}(z)} \qquad \forall z \in \phi_\rho((0,\infty)). \tag{8}$$

Therefore the map

$$h_p := \widetilde{f}_p \circ (f_p^\star)^{-1}$$

is increasing and concave. By Lemma B.2, every equal-price equalizing transfer endorsed by $W^\star$ at price p is also endorsed by $\widetilde{W}$ at that same price. Since p and $\widetilde{W}$ were arbitrary, $W^\star$ is ϕ_ρ-prioritarian.

Conversely, suppose a regular prioritarian social ordering is ϕ_ρ-prioritarian and is represented as

$$W(x) = \sum_i F(U(x_i))$$

for some increasing C^2 transform F. Since the ranking is ϕ_ρ-admissible, Lemma B.4 gives

$$A_F(u) := 1 + u\frac{F''(u)}{F'(u)} \leq a_\rho(u) \qquad \forall u \in Y. \tag{9}$$

First note that a_ρ is finite-valued. Indeed, for any fixed $p_0 \in \Delta$,

$$a_\rho(u) \leq a_\rho^0(u) \leq \rho\varepsilon(p_0, u) + \eta(p_0, u) < +\infty.$$

On the other hand, (9) gives the finite lower bound $A_F(u) \leq a_\rho(u)$, so $a_\rho(u) > -\infty$ for every $u \in Y$.

We claim that equality must hold in (9). It is enough to rule out strict inequality on the relative interior of Y, since equality then extends to endpoints by continuity whenever endpoints are included. Suppose, toward a contradiction, that for some u_1 in the interior of Y,

$$A_F(u_1) < a_\rho(u_1).$$

Set

$$\gamma := a_\rho(u_1) - A_F(u_1) > 0.$$

Because A_F is continuous and a_ρ is lower semicontinuous by definition, there exists an open interval $J \subset Y$ containing u_1 such that, for all $u \in J$,

$$A_F(u) \leq A_F(u_1) + \frac{\gamma}{4}, \qquad a_\rho(u) \geq a_\rho(u_1) - \frac{\gamma}{4}.$$

Let $\delta := \gamma/8$. Then, for every $u \in J$,

$$A_F(u) + 2\delta \leq A_F(u_1) + \frac{\gamma}{2} = a_\rho(u_1) - \frac{\gamma}{2} \leq a_\rho(u). \tag{10}$$

Choose a nonzero smooth function $q \geq 0$ supported in J, with $\|q\|_\infty \leq \delta$. Fix some base point $\bar{u} \in Y$, and define $\widetilde{F}$ by

$$\widetilde{F}'(u) := F'(u) \exp\left(\int_{\bar{u}}^u q(s)\, d\log s\right), \qquad \widetilde{F}(\bar{u}) := F(\bar{u}).$$

Then $\widetilde{F}$ is increasing and C^2, and

$$1 + u\frac{\widetilde{F}''(u)}{\widetilde{F}'(u)} = A_F(u) + q(u). \tag{11}$$

By (10), by the bound on q, and because q is supported in J,

$$1 + u\frac{\widetilde{F}''(u)}{\widetilde{F}'(u)} \leq a_\rho(u) \qquad \forall u \in Y.$$

Hence $\widetilde{F}$ is still ϕ_ρ-admissible by Lemma B.4.

Choose $\widehat{u} \in J$ such that $q(\widehat{u}) > 0$. Fix any $p \in \Delta$ and set

$$I_1 := e(p, \widehat{u}).$$

Define

$$\chi_p(I) := F(V(p, I)), \qquad \widetilde{\chi}_p(I) := \widetilde{F}(V(p, I)),$$

and

$$f_p := \chi_p \circ \phi_\rho^{-1}, \qquad \widetilde{f}_p := \widetilde{\chi}_p \circ \phi_\rho^{-1}.$$

Using formula (2), evaluated at $\widehat{u} = V(p, I_1)$, and using (11), we obtain

$$-\frac{I_1\chi_p''(I_1)}{\chi_p'(I_1)} - \left(-\frac{I_1\widetilde{\chi}_p''(I_1)}{\widetilde{\chi}_p'(I_1)}\right) = \frac{q(\widehat{u})}{\varepsilon(p, \widehat{u})} > 0.$$

Therefore, at $z_1 := \phi_\rho(I_1)$,

$$-\frac{f_p''(z_1)}{f_p'(z_1)} > -\frac{\widetilde{f}_p''(z_1)}{\widetilde{f}_p'(z_1)}.$$

By Lemma B.3, there exists a nearby equalizing transfer in ϕ_ρ-space that is endorsed by f_p but not by $\widetilde{f}_p$. Mapping back through ϕ_ρ^{-1}, this yields an equal-price equalizing transfer endorsed by W but not by the admissible ranking

$$\widetilde{W}(x) := \sum_i \widetilde{F}(U(x_i)).$$

This contradicts the least-inequality-aversion part of Axiom 3.8. Hence equality must hold in (9):

$$1 + u\frac{F''(u)}{F'(u)} = a_\rho(u) \qquad \forall u \in Y. \tag{12}$$

Since the left-hand side is continuous, a_ρ is continuous. Integrating (12) shows that there exist constants $\alpha \in \mathbb{R}$ and $\beta > 0$ such that

$$F(u) = \alpha + \beta F^\star(u).$$

This is the canonical form announced in the theorem.

It remains to check invariance with respect to the initial ordinal representation. Let $U' = T \circ U$, where $T : Y \to Y'$ is an increasing C^2 reparameterization. Define

$$\kappa(u) := \frac{d \log T(u)}{d \log u} > 0, \qquad \lambda(u) := \frac{d \log \kappa(u)}{d \log u}.$$

The transformed expenditure function is

$$e'(p,u') = e(p,T^{-1}(u')).$$

Writing $u' = T(u)$, one has $e'(p,T(u)) = e(p,u)$. Taking logarithms and differentiating with respect to $\log u$ gives

$$\varepsilon'(p,T(u))\kappa(u) = \varepsilon(p,u), \qquad \varepsilon'(p,T(u)) = \frac{\varepsilon(p,u)}{\kappa(u)}.$$

Differentiating once more in $\log u$ gives

$$\eta'(p,T(u))\kappa(u) = \eta(p,u) - \lambda(u), \qquad \eta'(p,T(u)) = \frac{\eta(p,u) - \lambda(u)}{\kappa(u)}.$$

Thus the raw curvature bounds satisfy

$$a^0_{\rho,U'}(T(u)) = \frac{a^0_{\rho,U}(u) - \lambda(u)}{\kappa(u)}.$$

Because T is a homeomorphism and κ, λ are continuous with $\kappa > 0$, taking lower-semicontinuous envelopes gives

$$a_{\rho,U'}(T(u)) = \frac{a_{\rho,U}(u) - \lambda(u)}{\kappa(u)}. \tag{13}$$

Let $F^\star_U$ and $F^\star_{U'}$ be the canonical transforms associated with U and U', normalized at corresponding base points. Since

$$\frac{d\log F^{\star\prime}_U(u)}{d\log u} = a_{\rho,U}(u) - 1$$

and

$$\frac{d\log F^{\star\prime}_{U'}(v)}{d\log v} = a_{\rho,U'}(v) - 1,$$

evaluating the second identity at $v = T(u)$ and multiplying by $\kappa(u)$ yields, using (13),

$$\frac{d\log F^{\star\prime}_{U'}(T(u))}{d\log u} = \kappa(u)\big(a_{\rho,U'}(T(u)) - 1\big) = a_{\rho,U}(u) - \lambda(u) - \kappa(u).$$

Moreover,

$$T'(u) = \frac{\kappa(u)T(u)}{u}, \qquad \frac{d\log T'(u)}{d\log u} = \lambda(u) + \kappa(u) - 1.$$

Hence

$$\frac{d}{d\log u}\log\big(F^{\star\prime}_{U'}(T(u))T'(u)\big) = a_{\rho,U}(u) - 1 = \frac{d\log F^{\star\prime}_U(u)}{d\log u}.$$

It follows that there exists $\beta > 0$ such that

$$F^{\star\prime}_{U'}(T(u))T'(u) = \beta F^{\star\prime}_U(u).$$

Integrating gives

$$F^\star_{U'}(T(u)) = \alpha + \beta F^\star_U(u)$$

for some $\alpha \in \mathbb{R}$. Thus the canonical representation is invariant up to a positive affine normalization, as required. $\square$

B.5 Proof of the money-metric characterization theorem 3.15

Proof. Fix the sequence $\tilde{p} = (p_n)_{n\in\mathbb{N}}$ and define

$$M(u) := \exp\left(\int_{u_0}^{u} \bar{\varepsilon}(s)\, d\log s\right), \qquad u \in Y,$$

so that the associated extended money-metric utility is $M_{\tilde{p}}(x) = M(U(x))$.

Let

$$F(u) := (\phi_\rho \circ M)(u).$$

By Theorem 3.11, $M_{\tilde{p}}$ is a representation of real income if and only if F has the canonical curvature profile,

$$1 + u\frac{F''(u)}{F'(u)} = a_\rho(u) \qquad \forall u \in Y. \tag{1}$$

We now compute the left-hand side of (1).

If $\rho \neq 0$, then

$$F(u) = \frac{M(u)^\rho}{\rho}.$$

Differentiating,

$$F'(u) = M(u)^{\rho-1} M'(u).$$

Taking logarithms and differentiating with respect to $\log u$ gives

$$\frac{d\log F'(u)}{d\log u} = (\rho - 1)\frac{d\log M(u)}{d\log u} + \frac{d\log M'(u)}{d\log u}.$$

Since

$$\frac{d\log M(u)}{d\log u} = \bar{\varepsilon}(u)$$

and

$$\frac{d\log M'(u)}{d\log u} = \bar{\varepsilon}(u) + \bar{\eta}(u) - 1, \qquad \bar{\eta}(u) := \frac{d\log \bar{\varepsilon}(u)}{d\log u},$$

we get

$$\frac{d\log F'(u)}{d\log u} = \rho\bar{\varepsilon}(u) + \bar{\eta}(u) - 1.$$

Thus

$$1 + u\frac{F''(u)}{F'(u)} = \rho\bar{\varepsilon}(u) + \bar{\eta}(u). \tag{2}$$

If $\rho = 0$, then $F(u) = \log M(u)$, and

$$F'(u) = \frac{M'(u)}{M(u)} = \frac{\bar{\varepsilon}(u)}{u}.$$

Therefore

$$\frac{d\log F'(u)}{d\log u} = \bar{\eta}(u) - 1,$$

so again

$$1 + u\frac{F''(u)}{F'(u)} = \rho\bar{\varepsilon}(u) + \bar{\eta}(u). \tag{3}$$

Combining the two cases, (2)-(3) give

$$1 + u\frac{F''(u)}{F'(u)} = \rho\bar{\varepsilon}(u) + \bar{\eta}(u) \qquad \forall u \in Y. \tag{4}$$

Assume first that conditions (1)-(2) of the theorem hold. Since $\bar{\varepsilon}$ is continuously differentiable and strictly positive, M is C^2, hence $F = \phi_\rho \circ M$ is C^2. By (4) and condition (2), F has the canonical curvature profile. By Theorem 3.11, $\phi_\rho \circ M$ is a canonical representation, hence $M_{\tilde{p}}$ is a representation of real income.

Conversely, suppose that $M_{\tilde{p}}$ is a representation of real income. Then, by Theorem 3.11, $F = \phi_\rho \circ M$ must satisfy (1). Combining (1) with (4) gives

$$\rho\bar{\varepsilon}(u) + \bar{\eta}(u) = a_\rho(u) \qquad \forall u \in Y.$$

Moreover, for the left-hand side to be well-defined as the curvature of the regular canonical representation, $\bar{\varepsilon}$ must be continuously differentiable. This proves the equivalence.

It remains to prove the final sufficient condition. Let

$$a_\rho^0(u) := \inf_{p\in\Delta}\{\rho\varepsilon(p,u) + \eta(p,u)\}$$

be the raw curvature bound. Assume condition (1) of the theorem, and suppose that

$$\tilde{p} \in \bigcap_{u\in Y} \arg\inf_{p\in\Delta}\{\rho\varepsilon(p,u) + \eta(p,u)\}$$

in the asymptotic sense that, for every $u \in Y$,

$$\rho\varepsilon(p_n,u) + \eta(p_n,u) \longrightarrow a_\rho^0(u).$$

Assume also that

$$\varepsilon(p_n,u) \to \bar{\varepsilon}(u), \qquad \eta(p_n,u) \to \bar{\eta}(u).$$

Passing to the limit gives

$$\rho\bar{\varepsilon}(u) + \bar{\eta}(u) = a_\rho^0(u) \qquad \forall u \in Y. \tag{5}$$

By condition (1), the left-hand side of (5) is continuous. Hence the raw curvature bound a_ρ^0 is continuous. Therefore it coincides with its lower-semicontinuous envelope a_ρ, and (5) is exactly condition (2). This proves the sufficient condition. □

B.6 Proof of the money-metric corollary 3.16

Proof. Fix $u \in Y$. By assumption,

$$\arg\sup_{p\in\Delta} \varepsilon(p,u) \subset \arg\inf_{p\in\Delta} \eta(p,u). \tag{1}$$

We claim that, because $\rho \le 0$, this implies

$$\arg\sup_{p\in\Delta} \varepsilon(p,u) \subset \arg\inf_{p\in\Delta} \{\rho\varepsilon(p,u) + \eta(p,u)\}. \tag{2}$$

Indeed, let $p^\star$ be any maximizer, or asymptotic maximizer, of $\varepsilon(\cdot,u)$. By (1), $p^\star$ is also a minimizer, or asymptotic minimizer, of $\eta(\cdot,u)$. Thus, for every $q \in \Delta$,

$$\varepsilon(p^\star,u) \ge \varepsilon(q,u), \qquad \eta(p^\star,u) \le \eta(q,u),$$

in the same asymptotic sense. Since $\rho \le 0$,

$$\rho\varepsilon(p^\star,u) \le \rho\varepsilon(q,u).$$

Adding the two inequalities gives

$$\rho\varepsilon(p^\star,u) + \eta(p^\star,u) \le \rho\varepsilon(q,u) + \eta(q,u),$$

again in the same asymptotic sense. This proves (2).

Now assume that

$$\tilde{p} \in \bigcap_{u\in Y} \arg\sup_{p\in\Delta} \varepsilon(p,u).$$

By (2), it follows that

$$\tilde{p} \in \bigcap_{u\in Y} \arg\inf_{p\in\Delta} \{\rho\varepsilon(p,u) + \eta(p,u)\}.$$

Let

$$a^0_\rho(u) := \inf_{p\in\Delta} \{\rho\varepsilon(p,u) + \eta(p,u)\}$$

be the raw curvature bound. Then, for every $u \in Y$,

$$\rho\varepsilon(p_n,u) + \eta(p_n,u) \longrightarrow a^0_\rho(u).$$

By definition of $\bar{\varepsilon}$ and by the additional assumption on the limit of the logarithmic derivatives,

$$\varepsilon(p_n,u) \to \bar{\varepsilon}(u), \qquad \eta(p_n,u) \to \bar{\eta}(u).$$

Passing to the limit gives

$$\rho\bar{\varepsilon}(u) + \bar{\eta}(u) = a^0_\rho(u) \qquad \forall u \in Y. \tag{3}$$

If condition (1) of Theorem 3.15 holds, namely if $\bar{\varepsilon}$ is continuously differentiable, then the left-hand side of (3) is continuous. Hence a^0_ρ is continuous and therefore coincides with the curvature envelope a_ρ. Thus condition (2) of Theorem 3.15 holds, and the extended money-metric utility $M_{\tilde{p}}$ is a representation of real income.

Finally, let $s \mapsto x_s$ be an absolutely continuous optimal consumption path associated with an absolutely continuous budget path $s \mapsto (p_s, I_s)$, and write $u_s := U(x_s)$. Since

$$U^*(x) = M_{\tilde{p}}(x) = M(U(x)), \qquad \frac{d \log M(u)}{d \log u} = \bar{\varepsilon}(u),$$

we have along the path

$$\frac{d \log U^*(x_s)}{ds} = \bar{\varepsilon}(u_s) \frac{d \log u_s}{ds}. \tag{4}$$

On the other hand, along an optimal path,

$$I_s = e(p_s, u_s) = p_s x_s.$$

Differentiating $\log e(p_s, u_s)$ with respect to s gives

$$\frac{d \log I_s}{ds} = \sum_\ell w_{\ell s} \frac{d \log p_{\ell s}}{ds} + \varepsilon(p_s, u_s) \frac{d \log u_s}{ds}, \tag{5}$$

where

$$w_{\ell s} := \frac{p_{\ell s} x_{\ell s}}{p_s x_s}$$

are expenditure shares. Likewise, differentiating $\log(p_s x_s)$ yields

$$\frac{d \log I_s}{ds} = \sum_\ell w_{\ell s} \frac{d \log p_{\ell s}}{ds} + \frac{p_s \dot{x}_s}{p_s x_s}. \tag{6}$$

Comparing (5) and (6), we obtain

$$\varepsilon(p_s, u_s) \frac{d \log u_s}{ds} = \frac{p_s \dot{x}_s}{p_s x_s} =: \mathrm{Div}(s). \tag{7}$$

Substituting (7) into (4),

$$\frac{d \log U^*(x_s)}{ds} = \frac{\bar{\varepsilon}(u_s)}{\varepsilon(p_s, u_s)} \mathrm{Div}(s).$$

Since $\tilde{p}$ asymptotically maximizes $\varepsilon(\cdot, u)$ for every u,

$$\bar{\varepsilon}(u) = \sup_{\tilde{p} \in \Delta} \varepsilon(\tilde{p}, u),$$

with the supremum understood in the same asymptotic sense as above. Therefore

$$\frac{d \log U^*(x_s)}{ds} = \frac{\sup_{\tilde{p} \in \Delta} \varepsilon(\tilde{p}, u_s)}{\varepsilon(p_s, u_s)} \mathrm{Div}(s).$$

Integrating from 0 to t yields the announced formula. □

B.7 Proof of the nh-CES growth formula

The expenditure function of the nh-CES system is

$$e(p,u) = \left(\sum_{i=1}^{L} \Omega_i p_i^{1-\sigma} u^{(1-\sigma)\epsilon_i}\right)^{\frac{1}{1-\sigma}}.$$

Let

$$w_i(p,u) := \frac{\Omega_i p_i^{1-\sigma} u^{(1-\sigma)\epsilon_i}}{\sum_j \Omega_j p_j^{1-\sigma} u^{(1-\sigma)\epsilon_j}}$$

be Hicksian budget shares. Then

$$\varepsilon(p,u) = \sum_i w_i(p,u)\epsilon_i =: E_w(\epsilon), \qquad \eta(p,u) = (1-\sigma)\frac{\mathrm{Var}_w(\epsilon)}{E_w(\epsilon)}. \tag{16}$$

If $\rho < 0$, the quantity

$$\rho\, E_w(\epsilon) + (1-\sigma)\frac{\mathrm{Var}_w(\epsilon)}{E_w(\epsilon)}$$

is minimized by the degenerate share vector concentrated on the largest elasticity ϵ_L: the first term is minimized when $E_w(\epsilon)$ is maximal, and the second term is always nonnegative and vanishes only at a degenerate share vector. Hence

$$a_\rho(u) = \rho\, \epsilon_L \qquad (\rho < 0).$$

Therefore the equality case in the canonical theorem is solved by

$$F(u) = \frac{u^{\rho\epsilon_L}}{\rho} = \phi_\rho(u^{\epsilon_L}),$$

so the real-income scale can be taken to be

$$U^*(x) = U(x)^{\epsilon_L}.$$

When $\rho = 0$, the same argument gives $a_0(u) = 0$, and any degenerate share vector is minimizing. Then the canonical transform is affine in $\log U$, so after the harmless normalization $\epsilon_L = 1$ one may take $U^*(x) = U(x)$.

Now consider an absolutely continuous Marshallian path $s \mapsto (p_s, x_s, I_s)$ and write $u_s := U(x_s)$. Since $I_s = e(p_s, u_s)$,

$$\mathrm{Div}(s) = \varepsilon(p_s, u_s)\frac{d\log u_s}{ds}. \tag{17}$$

If $\rho < 0$ and $U^* = U^{\epsilon_L}$, then

$$\frac{d\log U^*(x_s)}{ds} = \epsilon_L \frac{d\log u_s}{ds} = \frac{\epsilon_L}{\varepsilon(p_s,u_s)}\,\mathrm{Div}(s). \tag{18}$$

Thus the multiplicative correction factor is

$$\Lambda(p, I) = \frac{\epsilon_L}{\varepsilon(p, u)}, \qquad u = V(p, I). \tag{19}$$

It remains to rewrite *(19)* in terms of Marshallian income elasticities. By Shephard's lemma,

$$x_i(p, u) = \Omega_i p_i^{-\sigma} u^{(1-\sigma)\epsilon_i} e(p, u)^{\sigma}.$$

Along a Marshallian demand at fixed p, write $I = e(p, u)$. Differentiating $\log x_i$ with respect to $\log I$ yields

$$\varepsilon_i^I(p, I) := \frac{\partial \log x_i(p, I)}{\partial \log I} = \sigma + (1 - \sigma)\frac{\epsilon_i}{\varepsilon(p, u)}. \tag{20}$$

Because $\sum_i w_i \varepsilon_i^I = 1$, the weighted variance of Marshallian income elasticities is

$$\mathrm{Var}_w(\varepsilon^I) = \sum_i w_i(\varepsilon_i^I)^2 - 1 = \frac{(1 - \sigma)^2}{\varepsilon(p, u)^2}\,\mathrm{Var}_w(\epsilon). \tag{21}$$

Using *(16)* and the identity $d \log I = \varepsilon(p, u)\, d \log u$ at fixed prices, we obtain

$$\frac{d \log \varepsilon(p, u)}{d \log I} = \frac{\eta(p, u)}{\varepsilon(p, u)} = \frac{1 - \sigma}{\varepsilon(p, u)^2}\,\mathrm{Var}_w(\epsilon) = \frac{1}{1 - \sigma}\,\mathrm{Var}_w(\varepsilon^I). \tag{22}$$

Since $\sigma < 1$ and $\epsilon_L = \max_i \epsilon_i$, the budget shares converge to the most luxurious good along any fixed-price income expansion, so

$$\varepsilon(p, u) \longrightarrow \epsilon_L \qquad \text{as } I \to \infty.$$

Integrating *(22)* from I to $+\infty$ therefore gives

$$\log \frac{\epsilon_L}{\varepsilon(p, u)} = \frac{1}{1 - \sigma}\int_I^{+\infty} \mathrm{Var}_w(\varepsilon^I(p, z))\, d \log z.$$

Exponentiating yields the corrected closed form

$$\Lambda(p, I) = \exp\left(\frac{1}{1 - \sigma}\int_I^{+\infty} \mathrm{Var}_w(\varepsilon^I(p, z))\, d \log z\right). \tag{23}$$

Combining *(18)* and *(23)* proves the growth formula. □

B.8 Proof of the unique-profile characterization theorem 5.4

Proof. Fix the regular contour scale $\mathcal{U} : \mathcal{I}_N \to Y \subseteq \mathbb{R}_{++}$. For every preference relation R represented in the fixed profile, write

$$U_R^{\mathcal{U}}(x) := \mathcal{U}(I(x, R)), \qquad V_R^{\mathcal{U}}(p, m) := \max\{U_R^{\mathcal{U}}(x) : px \leq m\}.$$

Thus

$$e^{\mathcal{U}}(p, u, R) = \min\{px : U_R^{\mathcal{U}}(x) \geq u\}$$

is the expenditure function dual to $V_R^{\mathcal{U}}$. A regular $\mathcal{U}$-utilitarian social ranking is represented by

$$W_F(x) = \sum_{i=1}^{N} F\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

for some increasing C^2 function $F : Y \to \mathbb{R}$ with $F' > 0$. Define the local curvature index

$$A_F(u) := 1 + u\frac{F''(u)}{F'(u)}.$$

For a fixed pair (R, p), define the one-person indirect social-welfare function

$$\Gamma_{R,p}^F(m) := F\big(V_R^{\mathcal{U}}(p, m)\big).$$

Let $u = V_R^{\mathcal{U}}(p, m)$, so that $m = e^{\mathcal{U}}(p, u, R)$. Since

$$\frac{d \log e^{\mathcal{U}}(p, u, R)}{d \log u} = \varepsilon^{\mathcal{U}}(p, u, R), \qquad \frac{d \log \varepsilon^{\mathcal{U}}(p, u, R)}{d \log u} = \eta^{\mathcal{U}}(p, u, R),$$

one has

$$\frac{du}{dm} = \frac{u}{m\varepsilon^{\mathcal{U}}(p, u, R)}.$$

Therefore

$$(\Gamma_{R,p}^F)'(m) = F'(u)\frac{u}{m\varepsilon^{\mathcal{U}}(p, u, R)}.$$

Taking a logarithmic derivative with respect to m gives

$$\frac{m(\Gamma_{R,p}^F)''(m)}{(\Gamma_{R,p}^F)'(m)} = \frac{A_F(u) - \eta^{\mathcal{U}}(p, u, R)}{\varepsilon^{\mathcal{U}}(p, u, R)} - 1. \tag{1}$$

We first record the admissibility condition. Since

$$-\frac{m\phi_\rho''(m)}{\phi_\rho'(m)} = 1 - \rho,$$

the function

$$f_{R,p}^F := \Gamma_{R,p}^F \circ \phi_\rho^{-1}$$

is concave on $\phi_\rho(\mathbb{R}_{++})$ if and only if

$$-\frac{m(\Gamma_{R,p}^F)''(m)}{(\Gamma_{R,p}^F)'(m)} \geq 1 - \rho.$$

Using (1), this is equivalent to

$$A_F(u) \leq \rho\varepsilon^{\mathcal{U}}(p, u, R) + \eta^{\mathcal{U}}(p, u, R) \qquad \forall p \in \Delta,\ \forall u \in Y. \tag{2}$$

As in the common-preference proof, concavity of $f_{R,p}^F$, together with monotonicity, is equivalent to the requirement that all transfers satisfying $P(\phi_\rho)$ at price p and preference relation R be socially endorsed. Since ϕ_ρ-prioritarianism among equals only applies to pairs with the same preference relation, and since every preference relation in the fixed profile is held by at least two individuals, W_F is ϕ_ρ-admissible if and only if (2) holds for every preference relation represented in the fixed profile.

Sufficiency. Assume that $a_\rho^{\mathcal{U}}$ is finite-valued and continuous on Y. Fix $u_0 \in Y$, and define

$$A_\rho^{\mathcal{U}}(u) := \int_{u_0}^{u} a_\rho^{\mathcal{U}}(s)\, d\log s, \qquad F_\rho^{\mathcal{U}}(u) := \int_{u_0}^{u} \exp(A_\rho^{\mathcal{U}}(t))\, d\log t.$$

Then $F_\rho^{\mathcal{U}}$ is increasing and C^2, and

$$A_{F_\rho^{\mathcal{U}}}(u) = 1 + u\frac{(F_\rho^{\mathcal{U}})''(u)}{(F_\rho^{\mathcal{U}})'(u)} = a_\rho^{\mathcal{U}}(u). \tag{3}$$

Because $a_\rho^{\mathcal{U}} \le a_\rho^{\mathcal{U},0}$, (3) implies (2). Hence

$$W^\star(x) := \sum_{i=1}^{N} F_\rho^{\mathcal{U}}\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

is ϕ_ρ-admissible.

Let $\widetilde{W}$ be any other regular ϕ_ρ-admissible $\mathcal{U}$-utilitarian ranking, represented by an increasing C^2 function $\widetilde{F}$ with $\widetilde{F}' > 0$. By the admissibility condition (2),

$$A_{\widetilde{F}}(u) \le \rho\varepsilon^{\mathcal{U}}(p,u,R) + \eta^{\mathcal{U}}(p,u,R) \qquad \forall R, p, u.$$

Thus $A_{\widetilde{F}} \le a_\rho^{\mathcal{U},0}$. Since $A_{\widetilde{F}}$ is continuous, it also lies below the lower-semicontinuous envelope:

$$A_{\widetilde{F}}(u) \le a_\rho^{\mathcal{U}}(u) = A_{F_\rho^{\mathcal{U}}}(u) \qquad \forall u \in Y. \tag{4}$$

Fix a preference relation R represented in the fixed profile and a price vector $p \in \Delta$. Combining (1) for $F_\rho^{\mathcal{U}}$ and $\widetilde{F}$ with (4) gives

$$-\frac{(\widetilde{f}_{R,p})''(z)}{(\widetilde{f}_{R,p})'(z)} \ge -\frac{(f^\star_{R,p})''(z)}{(f^\star_{R,p})'(z)} \qquad \forall z \in \phi_\rho(\mathbb{R}_{++}),$$

where

$$f^\star_{R,p} := \Gamma_{R,p}^{F_\rho^{\mathcal{U}}} \circ \phi_\rho^{-1}, \qquad \widetilde{f}_{R,p} := \Gamma_{R,p}^{\widetilde{F}} \circ \phi_\rho^{-1}.$$

It follows that $\widetilde{f}_{R,p} \circ (f^\star_{R,p})^{-1}$ is increasing and concave. By Lemma B.2, every equal-price equalizing transfer between two individuals with preference relation R that is endorsed by $W^\star$ is also endorsed by $\widetilde{W}$. Since R, p, and $\widetilde{W}$ were arbitrary, $W^\star$ is least inequality averse among the regular ϕ_ρ-admissible $\mathcal{U}$-utilitarian rankings. Hence $W^\star$ is ϕ_ρ-prioritarian.

Necessity. Conversely, let W_F be a regular $\mathcal{U}$-utilitarian ranking that is ϕ_ρ-prioritarian. In particular, it is ϕ_ρ-admissible. By (2),

$$A_F(u) \le \rho\varepsilon^{\mathcal{U}}(p,u,R) + \eta^{\mathcal{U}}(p,u,R) \qquad \forall R, p, u.$$

Therefore

$$A_F(u) \le a_\rho^{\mathcal{U}}(u) \qquad \forall u \in Y, \tag{5}$$

because A_F is continuous. The envelope is finite-valued: it is bounded above by any fixed finite term $\rho\varepsilon^{\mathcal{U}}(p_0, u, R_0) + \eta^{\mathcal{U}}(p_0, u, R_0)$, and it is bounded below by the finite function A_F.

We show that equality must hold. Suppose first that strict inequality holds at some u_1 in the relative interior of Y:

$$A_F(u_1) < a_\rho^{\mathcal{U}}(u_1).$$

By continuity of A_F and lower semicontinuity of $a_\rho^{\mathcal{U}}$, there exist an open interval $J \subset Y$ containing u_1 and a number $\delta > 0$ such that

$$A_F(u) + 2\delta \le a_\rho^{\mathcal{U}}(u) \qquad \forall u \in J. \tag{6}$$

Choose a nonzero smooth function $q \ge 0$ supported in J, with $\|q\|_\infty \le \delta$, and define $\widetilde{F}$ by

$$\widetilde{F}'(u) := F'(u) \exp\left(\int_{\bar{u}}^{u} q(s)\, d\log s \right), \qquad \widetilde{F}(\bar{u}) := F(\bar{u})$$

for some $\bar{u} \in Y$. Then $\widetilde{F}$ is increasing and C^2, with

$$A_{\widetilde{F}}(u) = A_F(u) + q(u) \le a_\rho^{\mathcal{U}}(u) \qquad \forall u \in Y.$$

By (2), the ranking

$$\widetilde{W}(x) := \sum_i \widetilde{F}\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

is regular, ϕ_ρ-admissible, and $\mathcal{U}$-utilitarian.

Pick $\widehat{u} \in J$ such that $q(\widehat{u}) > 0$, choose a preference relation R represented in the fixed profile, and choose $p \in \Delta$. Let

$$m_1 := e^{\mathcal{U}}(p, \widehat{u}, R).$$

Using (1) at m_1 gives

$$-\frac{m_1 (\Gamma_{R,p}^{F})''(m_1)}{(\Gamma_{R,p}^{F})'(m_1)} > -\frac{m_1 (\Gamma_{R,p}^{\widetilde{F}})''(m_1)}{(\Gamma_{R,p}^{\widetilde{F}})'(m_1)}.$$

Equivalently, at $z_1 := \phi_\rho(m_1)$,

$$-\frac{(f_{R,p}^{F})''(z_1)}{(f_{R,p}^{F})'(z_1)} > -\frac{(f_{R,p}^{\widetilde{F}})''(z_1)}{(f_{R,p}^{\widetilde{F}})'(z_1)}.$$

Lemma B.3 yields a nearby equalizing transfer in ϕ_ρ-space that is endorsed by W_F but not by $\widetilde{W}$. Since each preference relation in the fixed profile is shared by at least two individuals, this transfer can be implemented between two individuals with preference relation R, at the common price p. This contradicts the least-inequality-aversion part of ϕ_ρ-prioritarianism. Hence equality holds on the interior of Y.

If Y contains an endpoint u_e, take a sequence u_n in the interior of Y converging to u_e. By lower semicontinuity and equality on the interior of Y,

$$a_\rho^{\mathcal{U}}(u_e) \le \liminf_{n\to\infty} a_\rho^{\mathcal{U}}(u_n) = \lim_{n\to\infty} A_F(u_n) = A_F(u_e).$$

Together with (5), this gives equality at u_e as well. Thus

$$A_F(u) = a_\rho^{\mathcal{U}}(u) \qquad \forall u \in Y. \tag{7}$$

In particular $a_\rho^{\mathcal{U}}$ is continuous. Integrating (7) gives constants $\alpha \in \mathbb{R}$ and $\beta > 0$ such that

$$F(u) = \alpha + \beta F_\rho^{\mathcal{U}}(u).$$

Therefore the social ranking is represented by

$$W_F(x) = \sum_{i=1}^{N} \big(F_\rho^{\mathcal{U}} \circ U_{R_i}^{\mathcal{U}}\big)(x_i),$$

up to a positive affine transformation. This proves the theorem. $\square$

B.9 Proof of the multi-profile characterization theorem 5.6

Proof. Let

$$\mathcal{I}^* := \{I(x,R) : x \in X,\ R \in \mathcal{R}^*\},$$

and fix a regular contour scale $\mathcal{U} : \mathcal{I}^* \to Y \subseteq \mathbb{R}_{++}$ with common image Y for every $R \in \mathcal{R}^*$. A regular $\mathcal{U}$-utilitarian multi-profile ranking is a family of profile-contingent rankings represented by a single increasing C^2 function $F : Y \to \mathbb{R}$, with $F' > 0$, such that for every $n \geq 1$, every profile $R^n = (R_1, \ldots, R_n) \in (\mathcal{R}^*)^n$, and every allocation $x \in X^n$,

$$W_{F,R^n}(x) = \sum_{i=1}^{n} F\big(U_{R_i}^{\mathcal{U}}(x_i)\big)$$

represents the ranking at profile R^n. Define again

$$A_F(u) := 1 + u\frac{F''(u)}{F'(u)}.$$

The calculation in the unique-profile proof applies to every $R \in \mathcal{R}^*$. Hence the family W_F is ϕ_ρ-admissible among equals on the whole multi-profile domain if and only if

$$A_F(u) \leq \rho\varepsilon^{\mathcal{U}}(p,u,R) + \eta^{\mathcal{U}}(p,u,R) \qquad \forall R \in \mathcal{R}^*,\ \forall p \in \Delta,\ \forall u \in Y. \tag{1}$$

Equivalently, $A_F \leq a_\rho^{\mathcal{U},0}$, and, since A_F is continuous, $A_F \leq a_\rho^{\mathcal{U}}$.

Sufficiency. Assume that $a_\rho^{\mathcal{U}}$ is finite-valued and continuous on Y, and construct $F_\rho^{\mathcal{U}}$ from $a_\rho^{\mathcal{U}}$ as in Theorem 3.11. Then

$$1 + u\frac{(F_\rho^{\mathcal{U}})''(u)}{(F_\rho^{\mathcal{U}})'(u)} = a_\rho^{\mathcal{U}}(u). \tag{2}$$

For every profile $R^n = (R_1, \ldots, R_n) \in (\mathcal{R}^*)^n$, define

$$W_{R^n}^{\star}(x) := \sum_{i=1}^{n} F_\rho^{\mathcal{U}}\big(U_{R_i}^{\mathcal{U}}(x_i)\big).$$

Because $a_\rho^{\mathcal{U}} \le a_\rho^{\mathcal{U},0}$, condition (1) holds for $F_\rho^{\mathcal{U}}$; hence $W^\star$ is ϕ_ρ-admissible among equals at every profile.

Now let $\widetilde{W}$ be any other regular ϕ_ρ-admissible $\mathcal{U}$-utilitarian multi-profile ranking, represented by $\widetilde{F}$. From (1),

$$A_{\widetilde{F}}(u) \le a_\rho^{\mathcal{U}}(u) = A_{F_\rho^{\mathcal{U}}}(u). \tag{3}$$

Fix any $R \in \mathcal{R}^*$ and any $p \in \Delta$. The curvature comparison induced by (3), exactly as in the unique-profile proof, implies that

$$\widetilde{f}_{R,p} \circ (f^\star_{R,p})^{-1}$$

is increasing and concave, where

$$f^\star_{R,p} := \Gamma^{F_\rho^{\mathcal{U}}}_{R,p} \circ \phi_\rho^{-1}, \qquad \widetilde{f}_{R,p} := \Gamma^{\widetilde{F}}_{R,p} \circ \phi_\rho^{-1}.$$

By Lemma B.2, every equal-price equalizing transfer between two individuals with preference relation R that is endorsed by $W^\star$ is also endorsed by $\widetilde{W}$. Since this is true for every $R \in \mathcal{R}^*$ and every $p \in \Delta$, $W^\star$ is least inequality averse among regular ϕ_ρ-admissible $\mathcal{U}$-utilitarian multi-profile rankings. Therefore $W^\star$ is ϕ_ρ-prioritarian.

Necessity. Conversely, let W_F be a regular $\mathcal{U}$-utilitarian multi-profile ranking that is ϕ_ρ-prioritarian. Since it is admissible, (1) gives

$$A_F(u) \le a_\rho^{\mathcal{U}}(u) \qquad \forall u \in Y. \tag{4}$$

The envelope is finite-valued: it is bounded below by A_F, and it is bounded above by any fixed term $\rho\varepsilon^{\mathcal{U}}(p_0, u, R_0) + \eta^{\mathcal{U}}(p_0, u, R_0)$, with $R_0 \in \mathcal{R}^*$ and $p_0 \in \Delta$.

Suppose that strict inequality holds at some u_1 in the interior of Y. As before, choose an interval $J \subset Y$, a number $\delta > 0$, and a nonzero smooth function $q \ge 0$ supported in J, with $\|q\|_\infty \le \delta$, such that

$$A_F(u) + 2\delta \le a_\rho^{\mathcal{U}}(u) \qquad \forall u \in J.$$

Define $\widetilde{F}$ by

$$\widetilde{F}'(u) := F'(u) \exp\left(\int_{\bar{u}}^{u} q(s)\, d\log s \right), \qquad \widetilde{F}(\bar{u}) := F(\bar{u}).$$

Then $A_{\widetilde{F}} = A_F + q \le a_\rho^{\mathcal{U}}$, so the multi-profile ranking represented at every profile by

$$\widetilde{W}_{R^n}(x) = \sum_{i=1}^{n} \widetilde{F}\big(U^{\mathcal{U}}_{R_i}(x_i)\big)$$

is regular, ϕ_ρ-admissible, and $\mathcal{U}$-utilitarian.

Pick $\widehat{u} \in J$ with $q(\widehat{u}) > 0$, choose any $R \in \mathcal{R}^*$, choose any $p \in \Delta$, and set

$$m_1 := e^{\mathcal{U}}(p, \widehat{u}, R).$$

The same local-curvature comparison as in the unique-profile proof gives, at $z_1 := \phi_\rho(m_1)$,

$$-\frac{(f^F_{R,p})''(z_1)}{(f^F_{R,p})'(z_1)} > -\frac{(f^{\widetilde{F}}_{R,p})''(z_1)}{(f^{\widetilde{F}}_{R,p})'(z_1)}.$$

Lemma B.3 gives an equal-price equalizing transfer, between two individuals with preference relation R, that is endorsed by W_F but not by $\widetilde{W}$. The profile (R, R) belongs to the multi-profile domain, so this transfer is admissible in the domain. This contradicts the least-inequality-aversion part of ϕ_ρ-prioritarianism. Hence equality holds on the interior of Y. The same lower-semicontinuity argument as in the unique-profile proof extends equality to any endpoint of Y, so

$$A_F(u) = a_\rho^{\mathcal{U}}(u) \qquad \forall u \in Y. \tag{5}$$

Thus $a_\rho^{\mathcal{U}}$ is finite-valued and continuous. Integrating (5) yields constants $\alpha \in \mathbb{R}$ and $\beta > 0$ such that

$$F(u) = \alpha + \beta F_\rho^{\mathcal{U}}(u).$$

Therefore, for every $n \geq 1$ and every $R^n = (R_1, \dots, R_n) \in (\mathcal{R}^*)^n$, the ranking at profile R^n is represented by

$$W_{R^n}(x) = \sum_{i=1}^{n} \big(F_\rho^{\mathcal{U}} \circ U_{R_i}^{\mathcal{U}}\big)(x_i),$$

up to a positive affine transformation of the profile-contingent welfare representation. This proves the theorem. □

B.10 Linking the construction of Theorem 3.11 to Kannai's result

The infimum over prices in this subsection is understood over nonzero supporting price vectors. When the main text restricts attention to strictly positive normalized prices, the resulting identities should be read in the corresponding boundary-limit sense.

When $\rho = 1$, our construction in Theorem 3.11 is closely related to Kannai's (Kannai 1977) construction of least concave utility functions. Kannai studies when a preference ordering can be *concavified* and, under smoothness assumptions, derives a formula for a minimally concave utility representation. Starting from an arbitrary utility function U, he looks for a function F such that $G := F \circ U$ is minimally concave. This requires the Hessian of G to be negative semidefinite everywhere. For a given utility level u, this yields the condition

$$\frac{F''(u)}{F'(u)} \leq -\frac{\sum_{i,j=1}^{m} U_{ij}(x)\,\xi_i\xi_j}{\big(\sum_{i=1}^{m} U_i(x)\,\xi_i\big)^2},$$

for every x such that $U(x) = u$ and every ξ such that $\sum_i U_i(x)\xi_i \neq 0$.

The author then shows that

$$\inf_{U(x)=u} \inf_{\xi} \left(-\frac{\sum_{i,j=1}^{m} U_{ij}(x)\,\xi_i\xi_j}{\big(\sum_{i=1}^{m} U_i(x)\,\xi_i\big)^2} \right) = \inf_{U(x)=u} \left(-\frac{S_{r(x)}(x)}{k(x)^2\, S^*_{r(x)-1}(x)} \right),$$

where:

- $S_i (= S_i(x))$ denotes the i-th elementary symmetric function of the eigenvalues of the quadratic form

$$\sum_{i,j=1}^{m} U_{ij}(x)\,\xi_i\xi_j,$$

- $r(x) - 1$ denotes the rank of that form restricted to

$$T_x := \Big\{ \xi \in \mathbb{R}^m : \sum_{i=1}^{m} \xi_i U_i(x) = 0 \Big\},$$

- S_i^* denotes the i-th elementary symmetric function of the eigenvalues of the restricted form on T_x,

- $k(x) := \|\nabla U(x)\|$.

Thus, constructing a least concave utility function amounts to choosing F so that, for all u,

$$\frac{F''(u)}{F'(u)} = \inf_{U(x)=u} \left(-\frac{S_{r(x)}(x)}{k(x)^2 \, S^*_{r(x)-1}(x)} \right).$$

Our approach takes a dual route, based on the indirect utility associated with G, denoted $\tilde{G}(p, I)$. Instead of studying the Hessian of G, we study the second derivative of $\tilde{G}$ with respect to income and ask for conditions ensuring that this derivative be nonpositive for every relevant price vector. For a given utility level u, this becomes

$$\frac{F''(u)}{F'(u)} \le \frac{e_{uu}(p, u)}{e_u(p, u)}.$$

Our dual approach therefore yields the equivalence

$$\inf_{p \in \Delta} \frac{e_{uu}(p, u)}{e_u(p, u)} = \inf_{U(x)=u} \left(-\frac{S_{r(x)}(x)}{k(x)^2 \, S^*_{r(x)-1}(x)} \right).$$

The following proposition and proof restate this equivalence directly in Kannai's framework.

Proposition B.5 (Link with Kannai's construction)**.** *Assume that U is C^2 in a neighborhood of the level set $\{x : U(x) = u\}$, and that the local hypotheses used by Kannai in Section 4 hold on that level set. Let U_i denote the derivative of U with respect to good i. Let $S_i (= S_i(x))$ denote the i-th elementary symmetric function of the eigenvalues of the quadratic form*

$$\sum_{i,j=1}^{m} U_{ij}(x) \, \xi_i \xi_j,$$

and let $r(x) - 1$ denote the rank of that form restricted to

$$T_x := \Big\{ \xi \in \mathbb{R}^m : \sum_{i=1}^{m} \xi_i U_i(x) = 0 \Big\}.$$

Denote by S_i^* the i-th elementary symmetric function of the eigenvalues of the restricted form on T_x, and set

$$k(x) := \|\nabla U(x)\|.$$

Define

$$G_K(u) := \inf_{U(x)=u} \left(-\frac{S_{r(x)}(x)}{k(x)^2\, S^*_{r(x)-1}(x)} \right).$$

Let also

$$\Gamma(u) := \Big\{ (p,x) : \ p \neq 0,\ U(x) = u,\ x \in \arg\min_{y \in X} \{ p \cdot y : U(y) \geq u \} \Big\}.$$

Then

$$G_K(u) = \inf_{(p,x)\in\Gamma(u)} \frac{e_{uu}(p,u)}{e_u(p,u)} = \inf_p \frac{e_{uu}(p,u)}{e_u(p,u)}.$$

Proof. Set

$$Q(x,\xi) := -\frac{\sum_{i,j=1}^m U_{ij}(x)\, \xi_i \xi_j}{\left(\sum_{i=1}^m U_i(x)\, \xi_i\right)^2}, \qquad \sum_i U_i(x)\xi_i \neq 0.$$

We will show that for every x such that $U(x) = u$,

$$\inf_\xi Q(x,\xi) = -\frac{S_{r(x)}(x)}{k(x)^2\, S^*_{r(x)-1}(x)} = \frac{e_{uu}(p,u)}{e_u(p,u)} \qquad \text{for every } (p,x) \in \Gamma(u).$$

Taking the infimum over x then yields the proposition.

Step 1. Computation of the inner infimum at a fixed point x.

Fix x such that $U(x) = u$. By translation and an orthogonal change of coordinates, we may assume that $x = 0$, that the tangent hyperplane to the level set $\{y : U(y) = u\}$ at 0 is

$$\{z \in \mathbb{R}^m : z_m = 0\},$$

and that the quadratic form

$$\sum_{i,j=1}^{m-1} U_{ij}(0)\, \zeta_i \zeta_j$$

restricted to $\{z_m = 0\}$ is diagonal. Then

$$U_i(0) = 0 \quad (1 \leq i \leq m-1), \qquad U_m(0) = k > 0,$$

and

$$U_{ij}(0) = 0 \quad (i \neq j,\ 1 \leq i,j \leq m-1).$$

Let $r = r(x)$. By definition of r,

$$U_{jj}(0) \neq 0 \quad (1 \leq j \leq r-1), \qquad U_{jj}(0) = 0 \quad (r \leq j \leq m-1).$$

Since U represents a convex preference ordering, the restricted quadratic form on the tangent space is negative semidefinite, hence

$$U_{jj}(0) < 0 \qquad (1 \leq j \leq r-1).$$

Moreover, Kannai's local condition III' implies

$$U_{jm}(0) = 0 \qquad (r \le j \le m-1).$$

Now fix $\xi = (\xi_1, \dots, \xi_m)$ with $\sum_i U_i(0)\xi_i \neq 0$. Since

$$\sum_{i=1}^{m} U_i(0)\xi_i = k\xi_m,$$

we must have $\xi_m \neq 0$. Set

$$t_j := \frac{\xi_j}{\xi_m} \qquad (1 \le j \le m-1).$$

Then

$$Q(0,\xi) = -\frac{\sum_{i,j} U_{ij}(0)\xi_i\xi_j}{(k\xi_m)^2}.$$

Using the diagonal form in the tangential directions and the vanishing of $U_{jm}(0)$ for $j \ge r$, we obtain

$$Q(0,\xi) = \frac{1}{k^2}\left(-U_{mm}(0) - 2\sum_{j=1}^{r-1} U_{jm}(0)t_j - \sum_{j=1}^{r-1} U_{jj}(0)t_j^2\right).$$

Because $U_{jj}(0) < 0$ for $1 \le j \le r-1$, the coefficient of each t_j^2 is $-U_{jj}(0) > 0$. Thus this is a strictly convex quadratic polynomial in $(t_1, \dots, t_{r-1})$, and its minimum is attained at

$$t_j^* = -\frac{U_{jm}(0)}{U_{jj}(0)}, \qquad 1 \le j \le r-1.$$

Substituting these values gives

$$\inf_{\xi} Q(0,\xi) = \frac{1}{k^2}\left(-U_{mm}(0) + \sum_{j=1}^{r-1} \frac{U_{jm}(0)^2}{U_{jj}(0)}\right) = -\frac{1}{k^2}\left(U_{mm}(0) - \sum_{j=1}^{r-1} \frac{U_{jm}(0)^2}{U_{jj}(0)}\right).$$

By Kannai's determinant identity (his equation (4.2)),

$$S_r(0) = \left(U_{mm}(0) - \sum_{j=1}^{r-1} \frac{U_{jm}(0)^2}{U_{jj}(0)}\right) S_{r-1}^*(0).$$

Hence

$$\inf_{\xi} Q(0,\xi) = -\frac{S_r(0)}{k^2 S_{r-1}^*(0)}.$$

Returning to the original point x, this gives

$$\inf_{\xi} Q(x,\xi) = -\frac{S_{r(x)}(x)}{k(x)^2\, S_{r(x)-1}^*(x)}.$$

Step 2. Identification with the expenditure ratio.

Fix now $(p, x) \in \Gamma(u)$. Since

$$e(\lambda p, s) = \lambda e(p, s) \qquad (\lambda > 0),$$

the ratio $e_{uu}(p, s)/e_u(p, s)$ is invariant under positive rescaling of p. We may therefore rescale p and keep the same adapted coordinates as above, with $x = 0$ and

$$p = e_m = (0, \ldots, 0, 1).$$

By the implicit function theorem, there exists a C^2 function

$$z_m = w(z_1, \ldots, z_{m-1}, s)$$

defined near $(0, u)$ such that

$$U(z_1, \ldots, z_{m-1}, w(z, s)) = s.$$

For fixed s, the graph of $w(\cdot, s)$ is the level set $\{U = s\}$. Since $p = e_m$, the expenditure problem is locally

$$e(p, s) = \min_{z \in \mathbb{R}^{m-1}} w(z, s).$$

At $(0, u)$, $z = 0$ is a local minimizer, so

$$w_j(0, u) = 0, \qquad 1 \le j \le m - 1.$$

Write $z = (z', z'')$, where

$$z' = (z_1, \ldots, z_{r-1}), \qquad z'' = (z_r, \ldots, z_{m-1}).$$

By construction,

$$w_{jj}(0, u) > 0 \quad (1 \le j \le r - 1), \qquad w_{jj}(0, u) = 0 \quad (r \le j \le m - 1),$$

and Kannai's condition III"' gives

$$w_{js}(0, u) = 0 \qquad (r \le j \le m - 1).$$

Let $z^*(s)$ be a local minimizing branch with $z^*(u) = 0$. The first-order conditions are

$$w_j(z^*(s), s) = 0, \qquad 1 \le j \le r - 1.$$

Differentiating at $s = u$ gives

$$w_{jj}(0, u)\, z_j^{*\prime}(u) + w_{js}(0, u) = 0, \qquad 1 \le j \le r - 1,$$

hence

$$z_j^{*\prime}(u) = -\frac{w_{js}(0, u)}{w_{jj}(0, u)}, \qquad 1 \le j \le r - 1.$$

Since

$$e(p, s) = w(z^*(s), s),$$

the envelope theorem yields

$$e_u(p,u) = w_s(0,u).$$

Differentiating once more, and using $w_j(0,u)=0$ together with the fact that the tangential Hessian is diagonal at $(0,u)$, we obtain

$$e_{uu}(p,u) = w_{ss}(0,u) + 2\sum_{j=1}^{r-1} w_{js}(0,u) z_j^{*\prime}(u) + \sum_{j=1}^{r-1} w_{jj}(0,u)\big(z_j^{*\prime}(u)\big)^2.$$

Substituting the formula for $z_j^{*\prime}(u)$, we obtain

$$e_{uu}(p,u) = w_{ss}(0,u) - \sum_{j=1}^{r-1} \frac{w_{js}(0,u)^2}{w_{jj}(0,u)}.$$

Now differentiate the identity

$$U(z, w(z,s)) = s.$$

At $(0,u)$, since $w_j(0,u)=0$, we get

$$U_m(0) = \frac{1}{w_s(0,u)} = k.$$

Hence

$$w_s(0,u) = \frac{1}{k}.$$

Differentiating the same identity again gives

$$U_{mm}(0) = -\frac{w_{ss}(0,u)}{w_s(0,u)^3}, \qquad U_{jm}(0) = -\frac{w_{js}(0,u)}{w_s(0,u)^2}, \qquad U_{jj}(0) = -\frac{w_{jj}(0,u)}{w_s(0,u)}.$$

Substituting these equalities into the formula for e_{uu}/e_u, and using $w_s(0,u)=1/k$, we get

$$\frac{e_{uu}(p,u)}{e_u(p,u)} = -\frac{1}{k^2}\left(U_{mm}(0) - \sum_{j=1}^{r-1}\frac{U_{jm}(0)^2}{U_{jj}(0)}\right).$$

By the calculation of Step 1,

$$\frac{e_{uu}(p,u)}{e_u(p,u)} = -\frac{S_r(0)}{k^2 S_{r-1}^*(0)} = \inf_\xi Q(0,\xi).$$

Returning to the original point x, this gives

$$\frac{e_{uu}(p,u)}{e_u(p,u)} = -\frac{S_{r(x)}(x)}{k(x)^2 S_{r(x)-1}^*(x)} = \inf_\xi Q(x,\xi).$$

Step 3. Conclusion.

Every x with $U(x) = u$ is a boundary point of the closed convex upper contour set

$$\{y \in X : U(y) \geq u\},$$

so it admits at least one supporting hyperplane. Therefore there exists $p \neq 0$ such that $(p, x) \in \Gamma(u)$. Conversely, every $(p, x) \in \Gamma(u)$ has $U(x) = u$. By Step 2, for every such pair,

$$\inf_{\xi} Q(x, \xi) = \frac{e_{uu}(p, u)}{e_u(p, u)}.$$

Hence

$$G_K(u) = \inf_{U(x)=u} \left(-\frac{S_{r(x)}(x)}{k(x)^2 \, S^*_{r(x)-1}(x)} \right) = \inf_{U(x)=u} \inf_{\xi} Q(x, \xi) = \inf_{(p,x)\in\Gamma(u)} \frac{e_{uu}(p, u)}{e_u(p, u)}.$$

Finally, the ratio $e_{uu}(p, u)/e_u(p, u)$ depends only on (p, u), not on the particular minimizing bundle x. Therefore

$$\inf_{(p,x)\in\Gamma(u)} \frac{e_{uu}(p, u)}{e_u(p, u)} = \inf_{p} \frac{e_{uu}(p, u)}{e_u(p, u)}.$$

This proves the proposition. $\square$